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Multi-Objective Optimization on Heat Transfer and Pumping Power of Combined Direct–Indirect Cooling System for PMSM Electric Motor

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Abstract

Efficient thermal management is essential for maintaining the reliability, durability, and power density of permanent magnet synchronous motors (PMSMs) used in electric vehicles (EVs). In this study, a combined direct–indirect cooling system (CDICS) was evaluated using multi-objective optimization to identify a constrained compromise between maximum component temperatures and ideal hydraulic pumping power. A three-dimensional CFD model was established to resolve conjugate heat transfer and oil–air two-phase flow within the proposed cooling configuration. The model was compared with winding temperature measurements obtained from motor dynamometer tests under two operating conditions, showing absolute CFD-measurement differences of 0.37 K and 0.50 K, respectively. Nozzle diameter (D_n), coolant volume flow rate (Q_{in}), and inlet oil temperature (T_{in}) were selected as the primary design and operating variables for optimizing the thermo-hydraulic performance of the proposed cooling system. Discrete-grid maximin space-filling design was used to generate 30 CFD design cases, which were used to develop artificial neural network (ANN) surrogate models for bounded interpolation within the investigated design space. The models showed close agreement with the available CFD data, with R^2 values of 0.9976–0.9994 for the six holdout cases. These values are interpreted as indicators of interpolation performance rather than evidence of universal generalization. The trained surrogate models were coupled with NSGA-II and TOPSIS to identify the optimal thermo-hydraulic compromise. The selected constrained compromise design within the investigated design space was D_n = 1.7 mm, Q_{in} = 16.59 LPM, and T_{in} = 15.0 °C. The selected condition yielded CFD-predicted maximum stator, winding, and housing temperatures of 107.5 °C, 109.8 °C, and 65.0 °C, respectively, with an ideal hydraulic pumping power of 20.0 W. Relative to the reconciled reference baseline, the maximum stator, winding, and housing temperatures differed nominally by <0.1 K, 0.3 K, and 1.5 K, respectively. Because these differences were smaller than the corresponding 1.98–2.29 K variations observed in the single-point laminar-SST comparison, all three temperature predictions were considered numerically comparable within the assessed model sensitivity rather than established thermal improvements. The CFD-estimated ideal hydraulic pumping power decreased by 9.0 W (31.0%). Because the baseline and selected conditions differed in both coolant flow rate and inlet temperature, and chiller and other auxiliary energy requirements were excluded, this value does not represent total cooling system energy savings.

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1. Introduction

Given concerns over climate change and greenhouse gas emissions, EVs have emerged as a viable alternative to conventional internal combustion engine vehicles [1–3]. As EV adoption continues to expand, the requirements for the performance, reliability, and durability of key powertrain components have become increasingly stringent [4–7]. Among various motor types, permanent magnet synchronous motors (PMSMs) are widely adopted because of their high efficiency, high torque density, and excellent dynamic response [8–11]. However, increased power density inevitably leads to substantial heat generation, particularly in the winding and stator regions, where copper loss and iron loss are concentrated. If this heat is not effectively dissipated, excessive temperature rise can deteriorate insulation materials, reduce magnetic performance, increase electrical resistance, and ultimately compromise motor efficiency, durability, and operational reliability [12–15]. Therefore, advanced thermal management strategies are essential for limiting critical component temperatures under high-load operating conditions, enabling further improvements in the performance and compactness of electric-drive systems while avoiding excessive auxiliary energy consumption [16–20].

Existing PMSM cooling strategies can be broadly classified into air cooling, indirect liquid cooling, and direct liquid/oil cooling [21–24]. Air cooling offers a simple and low-cost solution, but its limited heat transfer capacity makes it unsuitable for high-power-density traction motors operating under severe thermal loads [25–27]. Indirect liquid cooling systems, particularly water-cooling jackets, offer greater heat removal capability and proven manufacturing feasibility [28–33]. Nevertheless, heat generated in the windings and stator must pass through several solid layers before reaching the coolant, leading to substantial thermal resistance and persistent internal hotspots [34–36]. Direct oil cooling shortens this thermal path by bringing dielectric oil into direct contact with the stator and windings. Oil dripping, spray, and jet-impingement studies have consequently demonstrated improved local cooling [37–43]. However, their performance is highly sensitive to coolant distribution, near-wall oil coverage, drainage, pressure drop, and pumping requirements [44–46]. Combined direct–indirect cooling can exploit the complementary advantages of both approaches by cooling internal heat-generating regions and the housing simultaneously [47,48], but the shared hydraulic network introduces additional coupling between oil distribution, component temperatures, and hydraulic resistance.

Intelligent optimization methods have increasingly been introduced to overcome the computational limitations of conventional one-factor-at-a-time analyses. These methods generally combine design-of-experiments sampling, high-fidelity numerical simulations, surrogate modeling, evolutionary optimization, and multi-criteria decision-making. Li et al. [49] compared 13 surrogate models for an internal oil/external water PMSM cooling system and coupled the selected back propagation neural network with six multi-objective optimization algorithms. Wang et al. [50] integrated a convolutional neural network with particle swarm optimization to determine the dimensions of a motor cooling pipeline while considering cooling performance and pressure drop. For direct oil cooling, Im et al. [51] employed Kriging-based optimization to identify heat transfer coefficients for an oil-cooled motor thermal network under different flow and load conditions. Lee et al. [52] performed Pareto-based optimization of a U-shaped water jacket using six geometric variables, whereas Li et al. [53] combined Latin hypercube sampling, a radial basis function surro-

gate, k-fold cross-validation, and NSGA-II to optimize motor temperature, temperature uniformity, and pumping demand. Intelligent algorithms have also been used to calibrate the thermal network parameters of oil-cooled PMSMs for steady-state and transient predictions [54]. These studies demonstrate the efficiency of surrogate-assisted optimization. Nevertheless, most address indirect water-jacket geometry, thermal model calibration or a predetermined cooling path. Comparatively little attention has been paid to the multi-objective optimization of a combined direct–indirect oil cooling system in which the coolant naturally distributes between two hydraulically coupled branches.

References [47,48] established the feasibility and parametric behavior of combined direct–indirect oil cooling. Reference [47] examined nozzle diameter, total flow rate, and heat load using separately supplied direct and indirect cooling paths, whereas Reference [48] evaluated prescribed direct/indirect inlet flow ratios using a steady single-phase model. Reference [49] previously demonstrated surrogate-assisted multi-objective optimization for a different PMSM configuration combining an external water jacket and internal shaft-oil cooling. Accordingly, the contribution of the present study is not the individual ANN, NSGA-II, or TOPSIS algorithms.

The incremental engineering contribution is the optimization and quantitative assessment of a single-inlet CDICS in which oil distributes naturally between hydraulically coupled direct and indirect branches. The model additionally represents grid-scale oil–air distribution and drainage using VOF with temperature-dependent ATF properties. A new 30-case CFD dataset was generated across nozzle diameter, total flow rate, and inlet temperature, with four separate objectives representing stator, winding, and housing temperatures, as well as ideal hydraulic pumping power. Local winding-temperature predictions were compared with motor-dynamometer measurements at two operating conditions. At the selected design, the naturally established direct/indirect outlet-flow proportions were 54.21%/45.79%, the end-winding macroscopic coverage index was 0.998764, and the mass- and oil-side enthalpy-balance discrepancies were 0.125% and 2.03%, respectively. Therefore, the study extends previous parametric investigations by quantifying coupled-branch behavior and identifying a preference-selected thermo-hydraulic compromise within the prescribed design space.

The subsequent sections of this manuscript are structured as follows. Section 2 delineates the numerical methodology, including the computational model, governing equations, boundary conditions, data reduction procedure, mesh independence test, and validation of the numerical model. Section 3 presents the results and discussion, beginning with the single-parameter analysis of D_n, Q_{in}, and T_{in}, followed by ANN surrogate-model validation, Pareto optimization, TOPSIS-based selection of a constrained compromise within the investigated design space, and CFD verification of the optimized case. Finally, Section 4 summarizes the main conclusions and highlights the thermo-hydraulic trade-offs and limitations of the optimized CDICS for PMSM thermal management.

2. Numerical Method

2.1. Description of Computational Model

Figure 1 illustrates the numerical model of the permanent magnet synchronous motor (PMSM) equipped with the CDICS investigated in this study. The computational domain primarily consists of the stator, windings, and housing, which are the principal components responsible for heat generation, conduction, and dissipation in the motor. The rotor, shaft, and bearings were not included in the present CFD domain because the objective was to optimize the stator-side combined direct–indirect cooling structure under a prescribed high-load thermal condition. This simplification reduced computational complexity while enabling grid-resolved representation of oil injection, macroscopic oil–air distribution,

drainage, and conjugate heat transfer around the windings, stator, and housing [47,48]. Nevertheless, rotor-induced airflow and oil motion may influence oil entrainment, liquid-film redistribution, local wetting coverage, and convective heat transfer, particularly at high rotational speeds. Therefore, the present model should be interpreted as a stator-side thermo-hydraulic optimization model rather than a complete rotating machine flow model. The agreement between the predicted and measured winding temperature at 10,000 rpm supports only the local winding temperature response at the sensor location under the tested condition, but it does not independently validate the local oil distribution or demonstrate that rotational effects are negligible. A coupled rotating-domain model incorporating the rotor, shaft, bearings, and associated losses will be developed in future work to quantify these effects. The material properties assigned to the motor components are summarized in Table 1. The winding insulation was not modeled as a separate solid layer in the simplified motor geometry. Instead, the thermal resistance of the insulated winding assembly, including copper conductors, enamel coating, impregnation resin, slot insulation, and contact resistance, was accounted for by assigning equivalent anisotropic thermal properties to the winding region. Accordingly, the effective winding thermal conductivity was set to 10, 15, and 250 W/m·K in the radial, tangential, and axial directions, respectively. The lower radial and tangential conductivities represent the additional thermal resistance introduced by insulation and resin between adjacent conductors and toward the stator core, whereas the much higher axial conductivity reflects the preferential heat conduction along the copper conductors. Therefore, the simplified winding region should be interpreted as an equivalent insulated winding assembly rather than a pure copper winding.

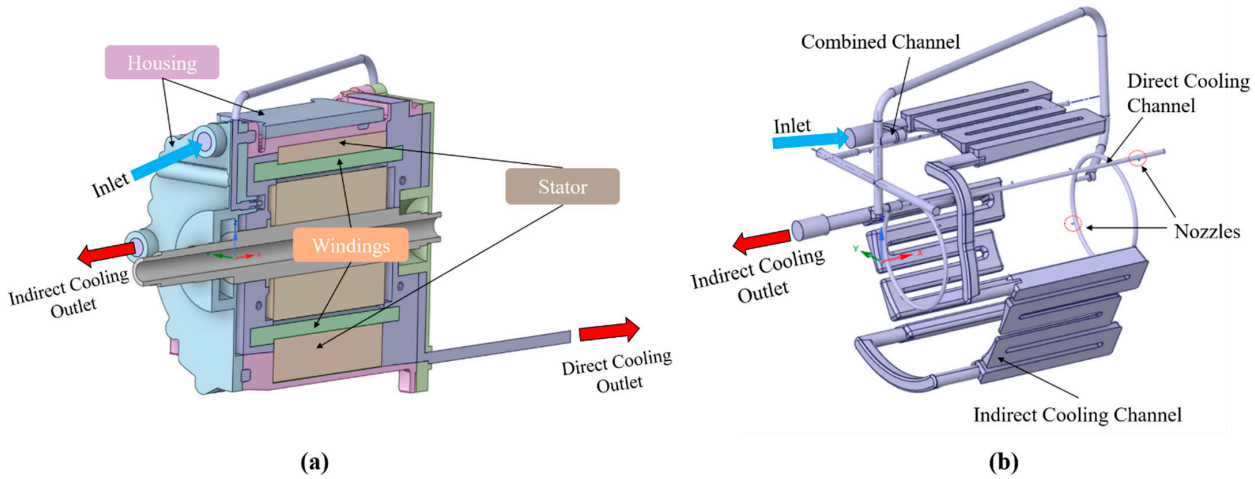


Figure 1. (**a**) Cross-sectional view of the integrated PMSM cooling configuration; (**b**) three-dimensional routing of the CDICS. The winding insulation is not shown as a separate layer because its effect is included in the equivalent anisotropic thermal properties of the winding region.

Table 1. Materials of electric motor components.

Material	Density (kg/m^3)	Specific Heat (J/kg·K)	Thermal Conductivity Radial/Tangential/Axial (W/m·K)
Stator (silicon steel)	8030	502.48	25/25/3
Winding (copper)	8978	381	10/15/250
Housing (cast aluminum)	2719	871	202.4

Note: The winding properties represent equivalent anisotropic properties of the insulated winding assembly; the insulation layer was not modeled as a separate solid domain.

The proposed cooling configuration consists of a single inlet, a combined distribution channel, a direct cooling section, and an indirect cooling section. The coolant oil first enters the cooling system through the inlet and is then delivered to the combined channel, where the flow is distributed into the direct and indirect cooling paths. In the direct cooling section, the coolant passes through internal supply passages and is injected into the motor interior through multiple nozzles. The injected oil directly impinges on, wets, and flows over the windings and stator surfaces, thereby removing heat from the main heat-generating regions. After heat exchange with these components, the coolant drains through the internal motor space and is discharged through the outlet. This direct-contact cooling mechanism is particularly important for suppressing localized temperature rise in the winding and stator regions.

In parallel, the indirect cooling section circulates the coolant through channels integrated within the housing. In this region, heat transferred from the internal motor components to the housing is removed by forced convection within the housing cooling passages. Therefore, the combined cooling structure enables simultaneous direct cooling of the main heat-generating regions and indirect cooling of the housing under a single-inlet configuration. This integrated design is expected to improve overall thermal management by reducing localized hot spots within the motor while limiting heat accumulation in the housing. Moreover, the combined flow arrangement provides a compact cooling architecture that can enhance heat dissipation without requiring separate coolant supply systems for the direct and indirect cooling regions.

The principal geometric dimensions of the direct- and indirect-cooling passages were obtained from the three-dimensional CAD model. The direct cooling network comprises two axial side manifolds and two end-region manifolds. Each side manifold has an internal diameter of approximately 4.0 mm and a straight length of approximately 223.9 mm, with eight injection nozzles arranged along each manifold. The end-region manifolds have the same internal diameter and centerline radii of approximately 49.0 and 50.5 mm, with three nozzles at each end. Thus, the direct cooling system contains 22 nozzles. The side- and end-nozzle axes are oriented at approximately 90° and 60°, respectively, relative to the motor axial direction. The nozzle diameter was varied from 1.1 to 1.7 mm, and the total direct-cooling outlet area was 9.99415×10^{-5} m^2. The indirect cooling network primarily comprises circular passages with a diameter of approximately 6.0 mm, positioned at a radial distance of approximately 108 mm from the motor axis. Its straight axial passage sections range from approximately 99.2 to 132.1 mm in length.

Because the coolant is injected into the internal motor space through nozzles in the direct cooling region, the flow field involves the coexistence of liquid oil and air. Accordingly, the VOF model was employed to represent the grid-scale oil–air phase distribution and drainage inside the motor. Because no separate wall-film formulation was used and surface-tension and contact-angle effects were excluded, the near-wall oil volume fraction was treated only as a macroscopic comparative indicator of coolant coverage. It should not be interpreted as a prediction of microscopic wetting, liquid-film thickness, or contact-line dynamics.

The main design and operating variables considered in this study were nozzle diameter, coolant volume flow rate, and inlet oil temperature. These parameters were selected because they directly affect both the heat transfer performance and hydraulic resistance of the combined cooling system. The nozzle diameter influences jet characteristics, local flow resistance, and oil distribution in the direct-cooling region. The coolant volume flow rate governs the heat capacity rate of the circulating oil and the strength of forced convection, while also affecting pressure loss and pumping power demand. The inlet oil temperature drives the convective heat transfer by establishing the thermal gradient between the coolant

oil and the solid surfaces. Simultaneously, it influences the oil's thermophysical properties, particularly viscosity, which directly governs the hydraulic pressure drop and the required pumping power. In this study, the D_n was varied from 1.1 to 1.7 mm, the Q_{in} from 10 to 20 LPM, and the T_{in} from 15 to 25 °C. The maximum temperatures of the stator, windings, and housing, together with the pressure drop and pumping power, were selected as the primary performance indicators for evaluating the thermo-hydraulic behavior of the proposed cooling system.

ATF-SP4 oil served as the working fluid throughout the numerical simulations. Since their density, specific heat capacity, thermal conductivity, and dynamic viscosity vary with temperature, these thermophysical properties were defined as temperature-dependent functions rather than constant values. The property data were implemented in ANSYS Fluent using a piecewise-linear interpolation method, allowing local fluid properties to be updated based on the cell temperature during the simulation. This treatment is important for accurately predicting the hydraulic and thermal behavior of oil-based cooling systems, particularly because viscosity strongly affects pressure drop and pumping power. Table 2 outlines the temperature-dependent thermophysical properties of the ATF-SP4 oil implemented in this simulation.

Table 2. Thermophysical properties of ATF-SP4 oil as a function of temperature [47,48].

Temperature (°C)	Density (kg/m^3)	Dynamic Viscosity (Pa·s)	Specific Heat Capacity (J/kg·K)	Thermal Conductivity (W/m·K)
0	856.5	0.10406	1831.24	0.139
15	846.1	0.07859	1886.54	0.137
40	829.5	0.01833	1978.69	0.136
80	803.3	0.00631	2126.14	0.133

Table 2 reports the ATF-SP4 properties over the complete 0–80 °C range implemented in Fluent. Across all simulated cases, the calculated oil-phase temperature ranged from 15 °C to 45.6 °C, as reported by global oil-domain temperature records. Therefore, the converged coolant temperatures remained within the tabulated property range. Density, dynamic viscosity, specific heat capacity, and thermal conductivity were evaluated by piecewise-linear interpolation between adjacent tabulated temperatures. For temperatures outside the specified range, the corresponding endpoint property value was retained rather than linearly extrapolated. No converged simulation required this endpoint clipping; consequently, property extrapolation did not influence the reported results.

2.2. Numerical Simulation Procedure and Boundary Conditions

The numerical procedure used to evaluate conjugate heat transfer and oil–air multiphase flow in CDICS is summarized in Figure 2. All simulations were performed using ANSYS Fluent 2025 R2 (Ansys Inc., Canonsburg, PA, USA). The computational domain included both the solid motor components and the coolant flow regions, allowing heat conduction in the stator, windings, and housing to be solved simultaneously with forced convection and oil–air interaction in the cooling passages.

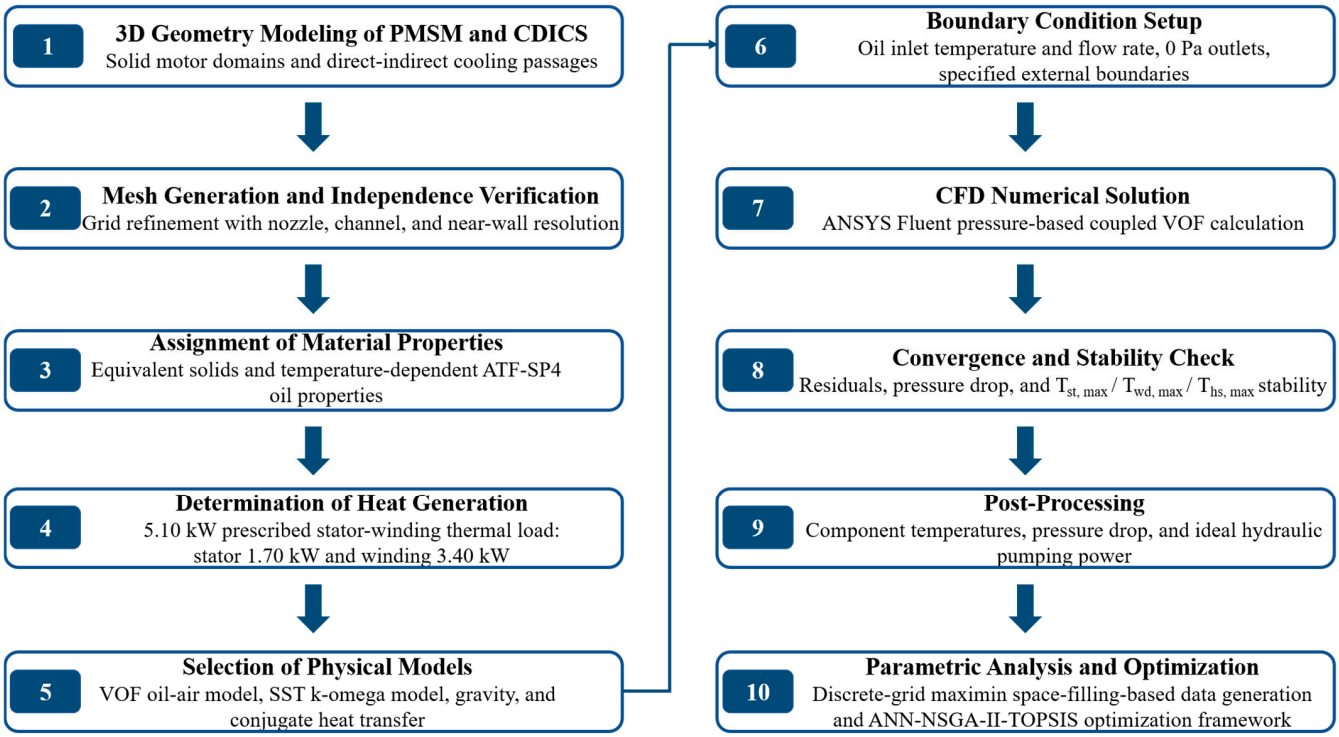


Figure 2. CFD simulation procedure for heat generation assignment, oil–air VOF multiphase flow calculation, conjugate heat transfer analysis, post-processing, and ANN-NSGA-II-TOPSIS optimization of the CDICS.

In the present simulation, heat generation was applied to the stator and winding regions to represent the major electromagnetic loss sources of the PMSM. The coolant-side heat absorption rate was determined by oil calorimetry under quasi-steady conditions, using the measured coolant mass flow rate, specific heat capacity, and the inlet-to-outlet temperature rise. This procedure yielded a nominal calorimetric load of $\dot{Q}_{heat}$ = 5.10 kW, which was prescribed as the modeled stator-winding thermal load. As detailed in Section 2.5, the adopted uncertainty of each standard-grade Type-K thermocouple was ±2.2 °C, giving a propagated uncertainty of ±3.11 K in the inlet-to-outlet temperature difference by the root-sum-square method. This uncertainty in temperature difference directly affects the inferred calorimetric load. Uncertainties associated with flow rate measurements and coolant property data may also contribute, but were not propagated separately. Therefore, 5.10 kW should be interpreted as a nominal calorimetric estimate rather than an exact measurement of the total motor loss.

The 5.10 kW calorimetric load was partitioned using component-level loss fractions obtained from Motor-CAD at the investigated operating point: 33.3% for the stator and 66.7% for the winding, giving 1.70 kW and 3.40 kW, respectively. Thus, Motor-CAD defined only the relative stator-winding allocation; the total imposed load remained the calorimetric estimate. These values were applied as uniform volumetric heat sources. The difference of approximately 0.64 kW between the estimated total electromechanical loss of 5.74 kW and the CFD-applied load of 5.10 kW is treated as an unresolved aggregate difference. It may include losses outside the modeled heat source domains, as well as uncertainties in calorimetric and electrical measurements and in the rounding of reported operating quantities. The available evidence does not permit this difference to be apportioned quantitatively among individual loss mechanisms. Thus, this treatment preserves the prescribed stator-winding heat input, rather than the total motor loss. Similar electromagnetic-loss-based thermal modeling approaches have been adopted in previous studies of electric motor thermal management [55]. Thus, the uniform volumetric heat generation assumption does

not imply that local electromagnetic losses are spatially uniform. Rather, it represents a homogenized component-level thermal load based on the experimentally determined total heat input and Motor-CAD-derived stator/winding loss allocation. For the homogenized winding region, the total winding loss was distributed over the complete equivalent insulated winding domain. Therefore, the winding region represents the combined thermal behavior of the copper conductors, insulation, impregnation material, and contact resistance, rather than that of pure copper alone. This heat source treatment enables a controlled evaluation of the CDICS under high thermal load conditions.

The solid domains were assigned their corresponding equivalent material properties. The ATF-SP4 oil was treated with temperature-dependent density, specific heat capacity, thermal conductivity, and dynamic viscosity. These properties were implemented in ANSYS Fluent using piecewise-linear interpolation, allowing the local coolant properties to be updated according to the cell temperature during the calculation. This treatment is important because oil viscosity varies strongly with temperature and directly affects the predicted pressure drop, flow distribution, and ideal hydraulic pumping power.

The anisotropic thermal conductivities were defined in the motor cylindrical coordinate system, consisting of the radial (r), tangential (θ), and axial (z) components. The stator core was modeled with (k_r, k_θ, k_z) = (25, 25, 3) W/(m·K) to account for the stacked steel laminations and interlaminar insulation layers. The homogenized winding domain used equivalent conductivities of (10, 15, 250) W/(m·K); here, the high axial value represents direct conduction along the continuous copper conductors, whereas the transverse values account for the thermal resistance of the wire insulation, impregnation material, and interconductor contact zones. The same conductivity tensor was applied directly to the end windings without local rotation along their curved conductor paths, a simplification that is recognized as a modeling limitation.

In the present model, the winding-stator and stator-housing interfaces were treated as perfectly bonded, with continuity of temperature and normal heat flux, and without a separately prescribed interfacial thermal contact resistance. The unresolved resistance associated with conductor insulation, impregnation resin, and internal winding contacts was incorporated into the equivalent anisotropic thermal properties of the homogenized winding region. Contact resistance arising from the stator-housing interference fit, surface roughness, or microscopic gaps was not modeled separately. All solid–fluid interfaces were represented as no-slip, thermally coupled walls under conjugate heat transfer, ensuring continuity of wall temperature and heat flux. These assumptions were applied consistently to all cases; nevertheless, their omission may affect the predicted absolute component temperatures.

The initial temperature of all solid and fluid regions was set to 25 °C. The steady-state VOF-based multiphase flow model was employed to simulate the oil–air two-phase flow in the direct cooling region [56,57]. The direct cooling cavity was initially filled with air, and the inlet phase was specified as pure ATF-SP4 oil with an oil volume fraction of 1.0. Gravity was included to represent the downward drainage and redistribution of oil after impingement. No separate wall-film model was used. Therefore, the VOF solution provides only the grid-scale oil–air volume-fraction distribution adjacent to solid surfaces and does not resolve the subgrid liquid-film thickness. Surface tension, wall adhesion, and contact-angle effects were not included. Consequently, the predicted near-wall oil distribution was used only for comparative assessment under consistently applied modeling assumptions. Absolute oil coverage and local convective heat transfer characteristics may change when these interfacial effects are included. Neither near-wall oil coverage nor detailed wetting behavior was independently validated experimentally.

For quantitative comparison, a macroscopic surface-coverage index was defined as

$$I_{COV} = \frac{1}{A_s}\int_{A_s} \alpha_{oil} dA \quad (1)$$

where α_{oil} is the wall-adjacent oil volume fraction, and A_s is the evaluated surface area. At the selected design, the end-winding cooling surface yielded I_{COV} = 0.998764, indicating nearly continuous macroscopic oil presence adjacent to this surface. This index should not be interpreted as a microscopic wetted-area fraction because the present VOF model does not resolve film thickness or contact-line behavior.

Conjugate heat transfer was applied at all solid–fluid interfaces, and coupled wall boundary conditions were used to ensure continuity of temperature and heat flux between the motor components and the coolant domain.

At the selected design condition D_n = 1.7 mm, Q_{in} = 16.59 LPM, and T_{in} = 15 °C, Reynolds numbers were estimated using $Re = \frac{\rho U D_h}{\mu}$, with ρ = 846.1 kg/m^3, μ = 0.07859 Pa·s, and $D_h = D_n$ for the circular nozzles. The nozzle-exit Reynolds numbers ranged from approximately 46 to 81. Based on the reported geometry and nozzle-flow extrema, the corresponding conservative Reynolds-number bounds were approximately 156–275 at the 4 mm side-manifold inlets, 59–103 at the 4 mm end-manifold inlets, and 153–269 at the direct cooling outlet, using an equivalent hydraulic diameter of 11.28 mm. For a 6 mm indirect cooling passage, the residual-flow estimate produced an upper-bound range of approximately 127–345; the actual local value depends on the naturally established branch split and parallel-channel distribution. These estimates indicate that the major passages remained nominally within the laminar Reynolds-number regime.

Accordingly, the SST $k-\omega$ model was not selected on the assumption of fully turbulent internal flow but was retained as a consistent RANS closure for possible localized shear, jet impingement, separation, and recirculation within the geometrically complex cooling network [58]. Near-wall resolution was evaluated using y^+. The area-weighted average y^+ values were 0.338, 0.509, and 0.660 for the stator, winding, and indirect channel walls, respectively, with corresponding maximum values of 1.332, 2.124, and 4.260. Thus, the principal heat transfer surfaces were predominantly resolved in the low-y^+ region.

To assess turbulence-closure sensitivity, an additional comparison between the laminar and SST $k-\omega$ models was conducted at the reference condition D_n = 1.70 mm, Q_{in} = 20.00 LPM, T_{in} = 20 °C, and a prescribed thermal load of 5.10 kW. Relative to the SST results, the laminar model predicted $T_{hs,\,max}$, $T_{st,\,max}$, and $T_{wd,\,max}$ values higher by 1.98 K, 2.25 K, and 2.29 K, respectively. The pressure drop was 0.59 kPa lower, while the direct- and indirect-branch flow fractions differed by only +0.25 and −0.25 percentage points, respectively. The macroscopic end-winding oil-coverage index was 99.88% under both models. Thus, the hydraulic distribution and grid-scale oil coverage were only weakly sensitive to the selected flow model at this operating point, whereas the component temperatures exhibited case-specific model-form sensitivity of approximately 2 K. Accordingly, temperature differences in comparable or smaller magnitude among candidate optimized designs should be interpreted with caution and not regarded as established thermal improvements. Because this comparison was conducted at only one operating point, it does not constitute a universal uncertainty bound or demonstrate preservation of the complete design ranking.

The coolant was supplied through a single common mass flow inlet as pure ATF-SP4 oil. The prescribed volumetric flow rates were converted to corresponding mass flow rates using the temperature-dependent density of ATF-SP4. At the direct cooling drain outlet, a pressure-outlet condition of 0 Pa gauge was imposed. The backflow volume fractions were specified as α_{air} = 1 and α_{oil} = 0, representing possible air ingress from the surrounding cavity during localized reverse flow. These values affect the solution only where outlet

backflow occurs. For the oil-filled indirect cooling outlet, the backflow condition was specified as pure oil.

The oil entering the common inlet was allowed to distribute naturally between the direct and indirect cooling branches according to the hydraulic resistance of each path. Therefore, the flow split was not prescribed separately. This boundary condition strategy enables the model to capture the coupled hydraulic behavior of the proposed CDICS.

As shown in Table 3, the direct and indirect outlets discharged 54.21% and 45.79% of the total outlet mass flow, respectively, demonstrating the naturally coupled flow distribution between the two cooling paths. The calculated mass imbalance was 0.125%. The net oil enthalpy transport was 4.997 kW, giving a 2.03% difference from the prescribed 5.10 kW stator-winding heat load. The reported enthalpies follow the Fluent reference state; therefore, their absolute values, including the negative inlet value, are not interpreted individually. Only the inlet–outlet enthalpy-flow difference is used in the energy balance.

Table 3. Branch-flow distribution and oil-side balance at the selected design.

Quantity	Inlet	Direct Outlet	Indirect Outlet
Mass flow rate (kg/s)	0.234000	0.126693	0.107015
Outlet-flow proportion (%)	-	54.21	45.79
Mass-weighted temperature (K)	288.150	299.831	298.774
Static enthalpy (J/kg)	−19,049.70	3239.67	1201.19

Initialization sensitivity, transient VOF behavior, rotor-induced flow, and material-specific wetting effects were not quantified. Consequently, the present analysis does not demonstrate that these simplifications preserve the design ranking under complete operation of the rotating machine. The reported ranking is restricted to the steady-state, rotor-excluded, stator-side VOF model. Future work will compare alternative phase initializations, perform transient VOF simulations for the reference and selected designs, incorporate a rotating rotor domain, and evaluate surface tension, contact angle, wall adhesion, and film behavior using material-specific oil-surface data.

No-slip boundary conditions were imposed on all solid walls. The external surfaces of the motor housing were treated as adiabatic to isolate the CDICS's internal heat removal capability from uncertain external influences, such as vehicle airflow, radiation, mounting conduction, and under-hood temperature. Accordingly, no external convective heat transfer coefficient, radiation boundary condition, or mounting thermal resistance was applied. Because the external housing boundaries were adiabatic, the prescribed 5.10 kW thermal load was transferred to the coolant at numerical steady state, apart from the numerical energy-balance residual. Consequently, differences in component temperature do not represent differences in total heat rejection. This controlled boundary condition was applied consistently across all cases, enabling a fair comparison of the investigated cooling designs. Accordingly, the predicted housing temperature represents the thermal response governed by the internal cooling system under the imposed adiabatic condition, rather than the complete vehicle-level thermal environment. The absolute housing temperature and even the relative design ranking may differ under realistic vehicle boundary conditions. Therefore, the present results should be interpreted as a controlled stator-side thermo-hydraulic assessment of the CDICS.

A pressure-based coupled solver was used to couple pressure and velocity. Second-order spatial discretization schemes were applied to the pressure, momentum, volume fraction, turbulence, and energy equations to improve numerical accuracy. Solution convergence was evaluated using scaled residual thresholds of 10^{-3} for the continuity, momentum, turbulence, and volume-fraction equations and 10^{-6} for the energy equation. In addition

to residual reduction, the pressure drop and the maximum temperatures of the stator, winding, and housing were monitored to confirm solution stability. Future studies will incorporate realistic vehicle airflow, mounting conduction, radiation, ambient temperature variation, and transient drive-cycle losses to further evaluate the full-system applicability of the optimized cooling structure.

All CFD simulations were performed in double precision using ANSYS Fluent 2025 R2 on a workstation equipped with a 32-core AMD Ryzen Threadripper PRO 7975WX processor and 512 GB of system memory. Using the selected 25.70-million-element production mesh, one representative CFD case required approximately 3 h of wall-clock time with all 32 CPU cores. Therefore, generating the 30-case CFD dataset required approximately 90 h of sequential wall-clock computation, equivalent to 2880 CPU core-hours. These estimates exclude the additional simulations conducted for mesh-sensitivity assessment, experimental comparison, independent ANN-CFD verification, and final design confirmation.

2.3. Data Reduction

The thermo-hydraulic performance of the CDICS was evaluated by post-processing CFD results. The maximum temperatures of the stator, winding, and housing were selected as the primary thermal performance indicators because these components represent the critical regions governing motor thermal reliability. The corresponding maximum temperatures, denoted as $T_{st,\,max}$, $T_{wd,\,max}$, and $T_{hs,\,max}$, were extracted from the calculated temperature fields of each solid domain.

The hydraulic performance of the cooling system was first assessed using the coolant pressure drop, which was calculated from the difference between the area-weighted average static pressures at the inlet and outlet [59,60]:

$$\Delta P_{coolant} = P_{inlet} - P_{outlet} \tag{2}$$

where $\Delta P_{coolant}$ is the coolant pressure drop, and P_{inlet} and P_{outlet} are the area-weighted average static pressures at the coolant inlet and outlet, respectively.

In addition to pressure drop, pumping power was calculated to quantify the ideal hydraulic power required to circulate the coolant oil through the combined cooling system. The hydraulic pumping power is defined as [59,60]:

$$P_{pump} = \Delta P_{coolant} \cdot Q_{in} \tag{3}$$

where P_{pump} is the pumping power, and Q_{in} is the volume flow rate of the coolant oil. Since the volume flow rate can be expressed as the product of inlet area and average inlet velocity, Equation (3) can also be written as [59,60]:

$$P_{pump} = \Delta P_{coolant} \cdot A_{in} \cdot u_{in} \tag{4}$$

where A_{in} and u_{in} are the inlet cross-sectional area and average inlet velocity, respectively. In this work, the reported pumping power represents the ideal hydraulic power required to overcome the coolant pressure loss, calculated from the CFD-predicted pressure drop and prescribed volumetric flow rate. This metric was selected because it directly reflects the hydraulic resistance introduced by the cooling structure and enables consistent comparison among different nozzle diameters and operating conditions without assuming a specific pump model. However, it does not represent the total auxiliary energy consumption of the complete vehicle cooling system. Pump efficiency, chiller power, heat exchanger performance, and coolant conditioning energy were not included. Therefore, P_{pump} should

be interpreted as an ideal hydraulic load indicator for comparative analysis, rather than the total energy consumption of the cooling system.

For the single-parameter analysis, $T_{st,\,max}$, $T_{wd,\,max}$, $T_{hs,\,max}$, and $\Delta P_{coolant}$ were compared to clarify the individual effects of D_n, Q_{in}, and T_{in}. For the subsequent multi-objective optimization, P_{pump} was used instead of pressure drop as the hydraulic objective because it directly reflects the energy penalty associated with coolant circulation. Thus, the optimization framework simultaneously considered thermal performance and ideal hydraulic pumping power.

2.4. Mesh Independence Test

A mesh-independence test was conducted to ensure that the numerical predictions were not significantly affected by the mesh resolution. The mesh sensitivity assessment was conducted at D_n = 1.50 mm, Q_{in} = 20.00 LPM, and T_{in} = 20 °C, with a prescribed stator-winding thermal load of 5.10 kW. Four mesh systems were generated with approximately 10.07, 11.91, 25.70, and 42.73 million elements. For each mesh, the $T_{st,\,max}$, $T_{wd,\,max}$, and $T_{hs,\,max}$, together with the $\Delta P_{coolant}$ across the cooling system, were compared, as shown in Figure 3.

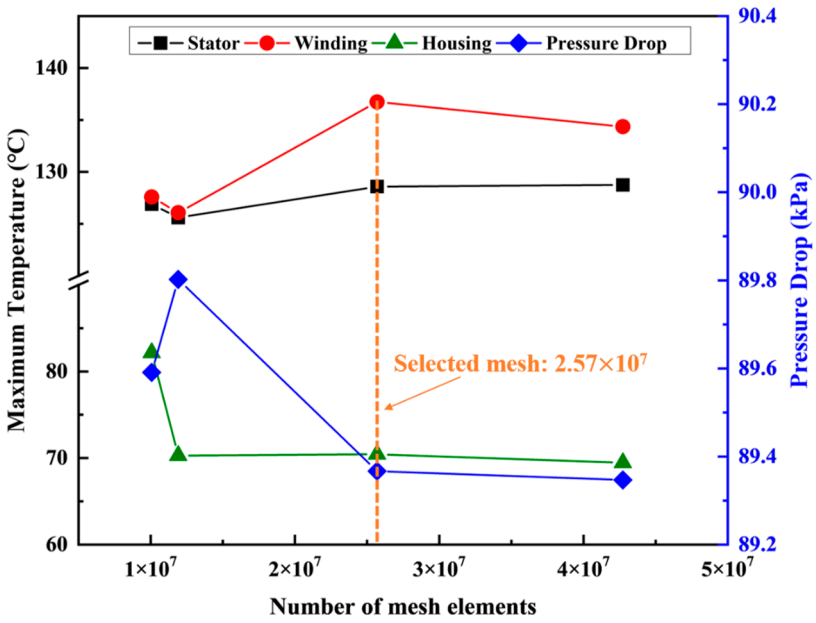


Figure 3. Mesh independence test based on maximum component temperatures and pressure drop.

As the mesh was refined from 10.07 to 11.91 million elements, $T_{wd,\,max}$ and $T_{st,\,max}$ decreased from 127.586 to 126.079 °C and from 126.899 to 125.605 °C, corresponding to absolute differences of 1.507 and 1.294 K, respectively. $T_{hs,\,max}$ decreased from 82.164 to 70.290 °C, representing a substantially larger difference of 11.874 K. These variations demonstrate that the coarsest mesh was insufficient to resolve the thermal gradients near the housing channels and solid–fluid interfaces.

Further refinement from 11.91 to 25.70 million elements increased $T_{wd,\,max}$ by 10.652 K, from 126.079 to 136.731 °C, and $T_{st,\,max}$ by 2.989 K, from 125.605 to 128.594 °C. Meanwhile, $T_{hs,\,max}$ changed by only 0.149 K, from 70.290 to 70.439 °C, and the pressure drop decreased by 0.435 kPa, from 89.802 to 89.367 kPa. The non-monotonic temperature behavior indicates that the local maxima are sensitive to oil impingement, oil coverage, and near-wall discretization.

For the final refinement from 25.70 to 42.73 million elements, the absolute differences in $T_{wd,\,max}$, $T_{st,\,max}$, and $T_{hs,\,max}$ were approximately 2.34, 0.15, and 0.98 K, respectively,

while the pressure drop difference was approximately 0.018 kPa. The normalized difference for each response X was defined relative to the finest-mesh result as:

$$\delta_X = \frac{|X_{25.70M} - X_{42.73M}|}{|X_{42.73M}|} \times 100\% \quad (5)$$

The corresponding normalized differences were 1.74%, 0.12%, 1.40%, and 0.02%, respectively. These case-specific results supported the selection of the 25.70-million-element mesh for comparative trend analysis. However, they are not interpreted as universal numerical-uncertainty bounds applicable to every investigated design.

A local nozzle-face element size of 0.15 mm was applied, providing approximately 10 elements across the nozzle diameter. Five inflation layers were generated using the Smooth Transition method, with a transition ratio of 0.272 and a growth rate of 1.2; therefore, the first-layer height was determined automatically rather than prescribed explicitly. The selected 25.70-million-element mesh had minimum, average, and maximum orthogonal-quality values of 0.00170, 0.8086, and 1.0, respectively. Only 25 cells, representing approximately 0.00010% of the mesh, had an orthogonal quality below 0.01, and these isolated cells were located in peripheral regions away from the nozzle and principal solid–fluid heat transfer interfaces. The area-weighted mean y^+ values for the stator, winding, and indirect channel walls were 0.338, 0.509, and 0.660, respectively, with corresponding maximum values of 1.332, 2.124, and 4.260.

Strict monotonic convergence was not achieved for all responses due to the complex, unstructured mesh, oil–air multiphase flow interface behavior, and the extraction of local maximum temperatures. Consequently, a formal Richardson-extrapolation-based GCI was not reported because the resulting apparent order would not provide a reliable uncertainty estimate. Instead, the deviation from the finest available mesh was reported transparently as a practical indicator of mesh sensitivity. The maximum deviation remained below 2%, while the pressure drop deviation was negligible. This mesh sensitivity was considerably smaller than the major parameter-driven variations, including the 15.1 K decrease in $T_{wd,\,max}$ and the 59.3 kPa (213.1%) increase in pressure drop produced by increasing flow rate in Section 3.1.2. Therefore, the selected mesh was considered adequate for identifying the dominant thermo-hydraulic trends at a substantially lower computational cost than the 42.73-million-element finest mesh. Nevertheless, temperature differences smaller than the corresponding mesh deviation, including the minor temperature changes associated with nozzle diameter and optimization, are interpreted as numerically comparable rather than definitive improvements. Figure 4 presents the computational grid configuration established for the subsequent numerical simulations.

2.5. Experimental Assessment of Local Winding Temperature

The local thermal response of the numerical model was assessed by comparing the predicted temperature at the winding surface sensor location with motor dynamometer measurements. The experimental assessment was restricted to the local winding temperature measured at the NTC sensor location. It was not intended to independently validate the maximum component temperatures, pressure drop, branch-flow distribution, or ideal hydraulic pumping power. To clarify the physical scale of the investigated motor, the main geometrical specifications of the PMSM used in the CFD model are summarized in Table 4. Only the macroscopic dimensions directly relevant to the thermal-fluid simulation are reported, including the motor length, stator dimensions, active stack length, pole-slot combination, shaft diameter, and air gap. Detailed proprietary electromagnetic design parameters, such as internal magnet dimensions, are not listed because the rotor and magnet assembly were not included as a heat-generating or fluid-contact domain in the

present stator-side CDICS model. These summarized dimensions are sufficient to define the thermal scale, heat transfer path, and cooling space configuration considered in the numerical analysis.

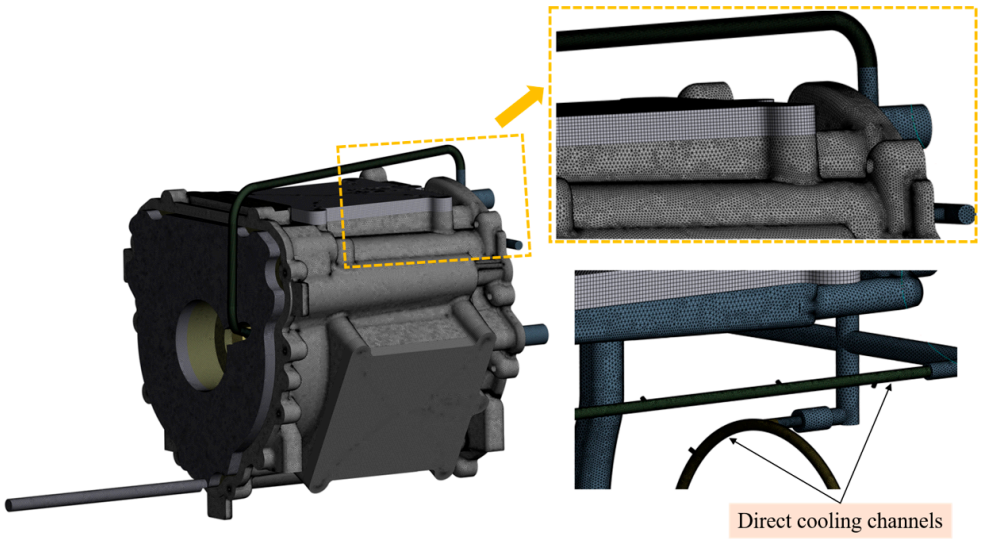


Figure 4. Computational mesh configuration of the integrated PMSM cooling system.

Table 4. Main geometrical specifications of the investigated PMSM.

Parameter	Value
Motor type	PMSM
Motor length	230 mm
Stator lamination outer diameter	200 mm
Stator bore diameter	134 mm
Active stator stack length	150 mm
Rotor stack length	150 mm
Number of poles	8
Number of slots	48
Shaft diameter	43 mm
Air gap	1.0 mm

The operating condition used for the numerical analysis is summarized in Table 5. The PMSM was operated at 10,000 rpm and 79.7 N·m, corresponding to a mechanical output power of 83.46 kW. This condition represents a high-speed, high-load operating point selected to evaluate the CDICS's cooling capability under a thermally demanding condition. Since high-power PMSM operation can generate a rapid temperature rise in the stator and winding regions, this operating point provides a meaningful basis for assessing the maximum temperature field and cooling performance. Therefore, the present simulation should be interpreted as a high-thermal-load assessment of the proposed cooling system.

The PMSM equipped with the CDICS was tested under two operating conditions with different heat loads and rotational speeds. The experimental measurements were then compared with the corresponding numerical predictions under identical operating and cooling conditions to evaluate the reliability of the developed CFD model. Figure 5 shows the schematic diagram of the motor dynamometer experimental setup used to validate the CDICS. The experimental facility consisted of an inverter, a PMSM, an oil reservoir, a pump, a chiller, a heat exchanger, and a closed-loop oil-circulation circuit. The motor operating speed and load were controlled using the inverter, while ATF-SP4 oil served as the dielectric cooling fluid. During operation, the coolant was stored in the

reservoir, cooled to the prescribed inlet temperature using the chiller and heat exchanger, and supplied to the motor cooling system by the pump. The coolant entered the motor through a single inlet and was subsequently distributed into the direct and indirect cooling sections through the combined channel. In the direct cooling section, the oil was supplied to the main heat-generating regions through multiple nozzles, including the windings and stator, and was discharged through the direct cooling outlet after heat exchange. In the indirect cooling section, the oil circulated through the housing cooling channels, removed heat transferred to the housing, and was discharged through the indirect cooling outlet. The discharged oil was then returned to the reservoir and recirculated through the cooling loop. K-type thermocouples were installed at representative locations in the oil circulation circuit to measure the inlet and outlet oil temperatures, while an NTC temperature sensor was attached to the winding surface to measure the local winding temperature. To demonstrate the reliability of the experimental setup and address the measurement uncertainties, the detailed instrumentation specifications, accuracy limits, and systematic limitations are summarized in Table 6. Figure 5 presents the experimental coolant circuit, the principal test equipment, and a photograph identifying the winding-embedded NTC sensor and its installation region. The winding temperature was selected as the representative validation parameter because it is one of the most thermally critical quantities in the motor. For consistency, the CFD-predicted winding temperature was extracted at the corresponding sensor location. The measured winding temperature was then compared with the CFD prediction to evaluate the accuracy of the numerical thermal model. This agreement supports the numerical model's ability to reproduce the winding's thermal response under the operating conditions investigated. Nevertheless, the stator temperature, housing temperature, pressure drop, branch-flow distribution, and ideal hydraulic pumping power were not independently validated experimentally and should therefore be interpreted as CFD-predicted indicators for comparative thermo-hydraulic analysis. Therefore, this comparison assesses only the local winding temperature response and should not be interpreted as validation of maximum component temperatures or hydraulic quantities. Future experimental work will include additional measurements of multi-location temperature, pressure drop, branch-flow, and pumping power to validate the complete CDICS behavior.

Each experimental condition was repeated three times. Steady state was defined as a winding temperature variation corresponding to $|dT_{wd}/dt| < 0.1$ K/min over a continuous 10 min interval. The reported winding temperatures represent the mean values from the three runs, and their repeatability is reported as the standard deviation, which remained below 0.15 K across all tested cases.

Table 5. Operating conditions and loss distribution of the investigated PMSM.

Parameter	Value
Rotational speed	10,000 rpm
Torque	79.7 N·m
Electrical input power	89.20 kW
Motor efficiency	93.57%
Mechanical output power	83.46 kW
Estimated total electromechanical loss	5.74 kW
Operating classification	High-speed, high-load condition
CFD-applied stator-winding heat load	5.10 kW
Stator heat load	1.7 kW
Winding heat load	3.4 kW
Difference between estimated total loss and CFD-applied thermal load	0.64 kW

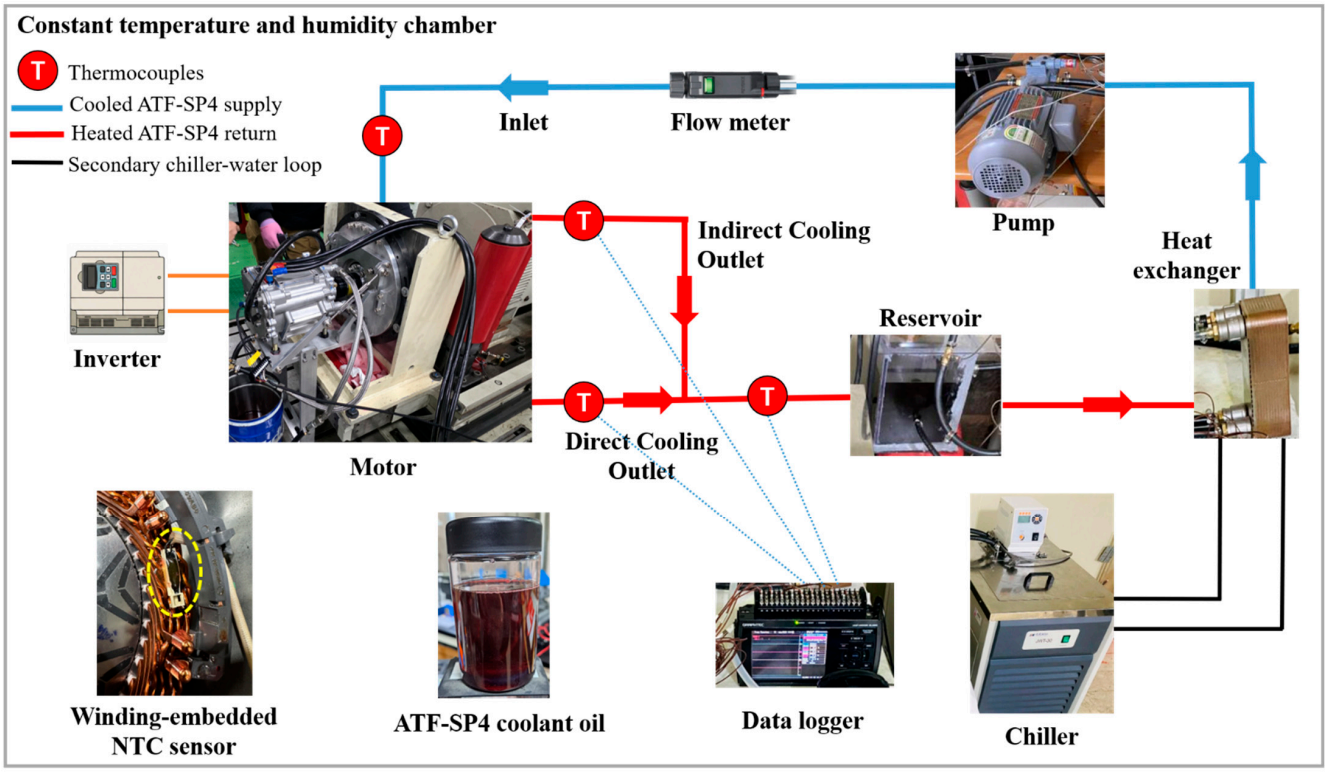


Figure 5. Experimental test setup and coolant-circuit connection diagram for the CDICS.

Table 6. Instrumentation and measurement performance information for the experimental comparison.

Quantity	Instrument	Accuracy/Uncertainty	Limitation
Local winding temperature	NTC sensor, R_{25} = 16.8 kΩ and B = 2230 K (Murata Manufacturing Co., Ltd., Kyoto, Japan)	Maximum conversion-model deviation: ±1.80 °C over 50.55–76.27 °C	Mounting/contact uncertainty not independently quantified
Inlet/outlet oil temperature	Standard-grade Type-K thermocouples (MISUMI Group Inc., Tokyo, Japan)	±2.2 °C per sensor; propagated ΔT_{oil} uncertainty: ±3.11 K by RSS	No individual sensor/logger calibration certificate was available; the stated value is based on the standard-grade thermocouple limit
Total inlet flow rate	Keyence FD-XA1, FD-XS20 and FD-XC20R1 (Keyence Corporation, Osaka, Japan)	Rated range: 0–15 LPM; conservative repeatability: ±0.015 LPM	Measures total inlet flow only, not branch flows

Two validation cases were selected to assess the numerical model's prediction accuracy under different thermal load conditions. Case 1 corresponded to a heating power of 2.6 kW and a rotational speed of 2257 rpm, whereas Case 2 corresponded to a heating power of 5.1 kW and a rotational speed of 10,000 rpm. Neither the equivalent anisotropic winding properties nor the heat source allocation was adjusted using the two experimental temperature comparisons. These quantities were fixed before comparing the experimental results with the CFD predictions. Therefore, the 3.4 kW winding and 1.7 kW stator heat inputs represent an a priori modeling allocation of the prescribed 5.1 kW stator-winding thermal load, rather than values fitted to the NTC measurements. The NTC results were used only for an independent posterior comparison at the corresponding sensor location.

For Case 2 validation, the measured mechanical output power and motor efficiency were 83.46 kW and 93.57%, respectively. Accordingly, the electrical input power was $P_{in} = P_{out}/\eta$ = 89.20 kW, yielding a total motor loss of 5.74 kW. Because volumetric heat sources were applied only to the stator and winding domains, a nominal total load of 5.10 kW was prescribed, comprising 3.40 kW in the winding and 1.70 kW in the stator. The two modeled heat source values were held constant across all parametric simulations and were not adjusted based on the resulting component temperatures. The approximately 0.64 kW difference from the estimated total electromechanical loss was not applied to the CFD model. This difference is not entirely attributable to particular unmodeled loss mechanisms; it may reflect a combination of losses outside the modeled heat source domains, calorimetric and electrical measurement uncertainties, and rounding of the reported quantities. Thus, the applied 5.10 kW refers to the specific stator-winding heat load rather than the motor's total electromechanical loss. To clarify the discrepancy raised during the validation process, a comprehensive energy and loss balance for validation Case 2 is summarized in Table 7. For both cases, the inlet oil temperature and coolant volume flow rate were maintained at 25 °C and 15 LPM, respectively, as summarized in Table 8. This validation strategy enabled the numerical model to be assessed under the same cooling conditions but different heat-generation and rotational-speed levels.

Table 7. Energy and loss balance for validation Case 2.

Quantity	Calculation or Allocation	Power (kW)
Mechanical output	Reported operating value	83.46
Electrical input	(83.46/0.9357)	89.20
Total motor loss	(89.20–83.46)	5.74
Winding heat applied in CFD	Prescribed allocation	3.40
Stator heat applied in CFD	Prescribed allocation	1.70
Total modeled heat load	(3.40 + 1.70)	5.10
Difference between estimated total loss and CFD-applied thermal load	(5.74–5.10)	0.64

Table 8. Experimental conditions of oil cooling system for electric motor.

	Heating Power (kW)	Speed (rpm)	Inlet Oil Temperature (°C)	Volume Flow Rate (LPM)
Case 1	2.6	2257	25	15
Case 2	5.1	10,000	25	15

The NTC sensor was mounted on the accessible outer winding surface, whereas the CFD-predicted maximum winding temperature occurred internally, within regions shielded from the direct oil spray. For consistency, the numerical temperature used in the experimental comparison was extracted from the computational cells corresponding to the exact NTC sensing location, rather than from the predicted hotspot. The differences between the local sensor-position temperature and the predicted maximum winding temperature were 16.37 °C and 37.94 °C for Cases 1 and 2, respectively. Consequently, the comparison provides independent evidence for the local winding-temperature response but does not experimentally validate the predicted winding hotspot. Multi-location winding temperature measurements will be included in future work.

Figure 6 compares the experimentally measured winding temperatures with the corresponding numerical predictions. The nominal differences between the CFD predictions and experimental data were 0.37 K and 0.50 K for Cases 1 and 2, respectively. Importantly, both

deviations fall well within the conservative uncertainty band of ±1.80 K associated with the NTC conversion model over the tested temperature range. Therefore, the CFD predictions are considered consistent with the local winding temperature measurements within the limits of experimental uncertainty. Crucially, this agreement should not be interpreted as a claim of sub-1% absolute model accuracy, nor does it constitute a direct validation of the maximum winding hotspot temperature. In addition, the maximum stator and housing temperatures, pressure drop, direct/indirect branch-flow distribution, and ideal hydraulic pumping power were not independently validated experimentally and should therefore be interpreted as CFD-predicted quantities for comparative thermo-hydraulic analysis within the scope of the present numerical model. Furthermore, the experimental comparison does not independently validate rotor-induced flow, external heat rejection, near-wall oil coverage, wall-film behavior, or surface-tension, wall-adhesion, and contact-angle effects.

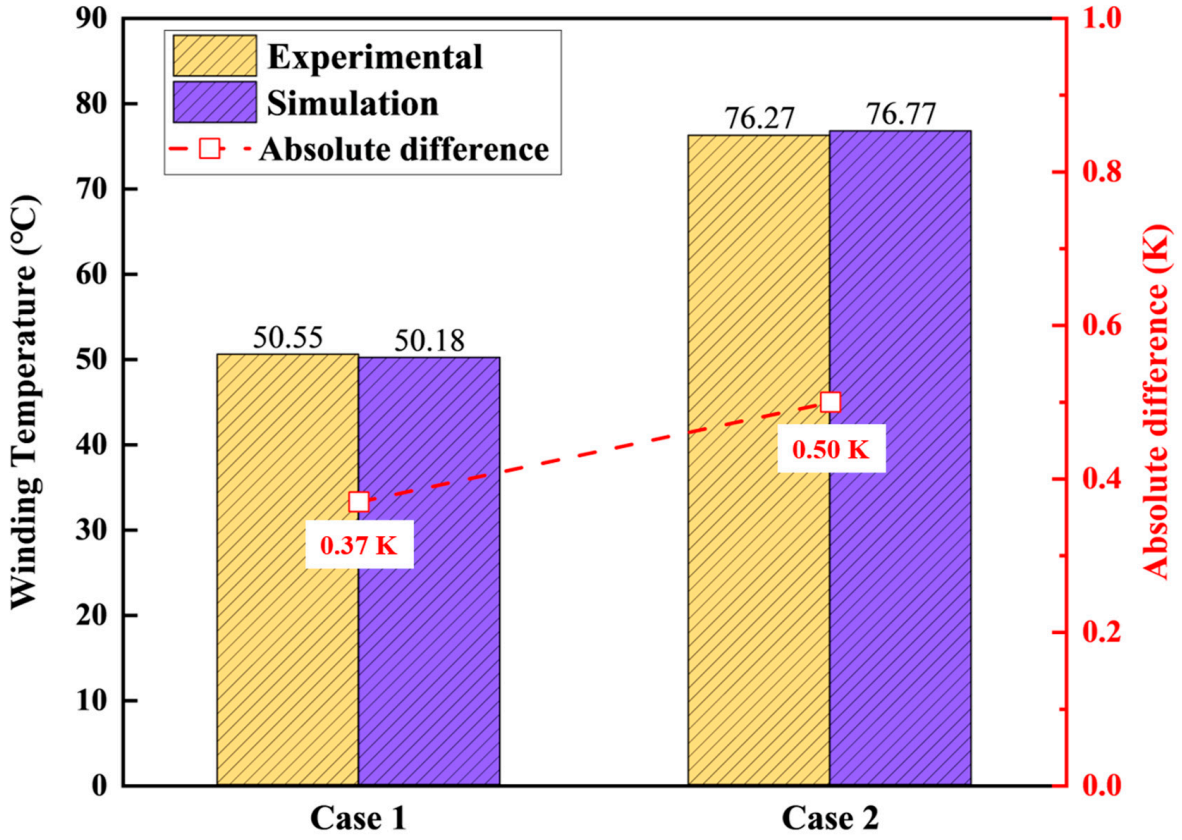


Figure 6. Comparison of measured and CFD-predicted local winding temperatures and their absolute differences at the NTC sensor location.

3. Results and Discussion

3.1. Single-Parameter Analysis

This section investigates the effects of key geometric and operating parameters on the thermo-hydraulic performance of the CDICS using a one-factor-at-a-time approach. The examined parameters include nozzle diameter (D_n), coolant volume flow rate (Q_{in}), and inlet oil temperature (T_{in}). For each parameter, the $T_{wd,\,max}$, $T_{st,\,max}$, and $T_{hs,\,max}$ together with the overall $\Delta P_{coolant}$ across the cooling circuit, were evaluated. This analysis quantitatively identifies the relative influence of each parameter on thermal performance and hydraulic resistance, thereby providing a physical basis for the subsequent multi-objective optimization.

Although several geometric parameters, such as channel thickness, channel width, and the number of cooling channels, can also affect the CDICS performance, these parameters

were not selected as optimization variables in the present study because they define the basic housing-integrated cooling architecture. Changing them would require a complete redesign of the motor housing, manifold layout, mechanical spacing, and manufacturing configuration. Therefore, they were fixed in accordance with the baseline CDICS design to maintain a consistent motor structure and isolate the effects of the most practically adjustable thermo-hydraulic variables. In contrast, nozzle diameter, inlet flow rate, and inlet oil temperature directly control jet momentum, wetting coverage, convective heat removal, and hydraulic loss, while requiring less modification to the overall motor structure. Accordingly, the present work focuses on these three dominant design and operating parameters as a first-stage optimization of the proposed CDICS.

3.1.1. Effect of Nozzle Diameter (D_n)

Nozzle diameter is a key geometric parameter that governs both the oil-spray characteristics in the direct cooling region and the hydraulic resistance of the flow passages. To isolate its influence, the inlet oil temperature and volumetric flow rate were maintained at 20 °C and 20 LPM, respectively. Numerical simulations were then conducted for nozzle diameters of 1.1, 1.3, 1.5, and 1.7 mm to evaluate the associated thermal and hydraulic responses.

Figure 7 presents the variations in the $T_{wd,\,max}$, $T_{st,\,max}$, and $T_{hs,\,max}$, together with the $\Delta P_{coolant}$ across the cooling system, for different nozzle diameters. As the D_n increased from 1.1 to 1.7 mm, the $T_{st,\,max}$ decreased from 108.037 °C to 107.512 °C, corresponding to a 0.5 K reduction. The $T_{wd,\,max}$ and $T_{hs,\,max}$ exhibited only minor variations, with overall reductions of 0.3 K and 0.1 K, respectively. Notably, small non-monotonic fluctuations occurred at intermediate diameters, indicating that nozzle diameter had a limited influence on thermal performance under the investigated conditions.

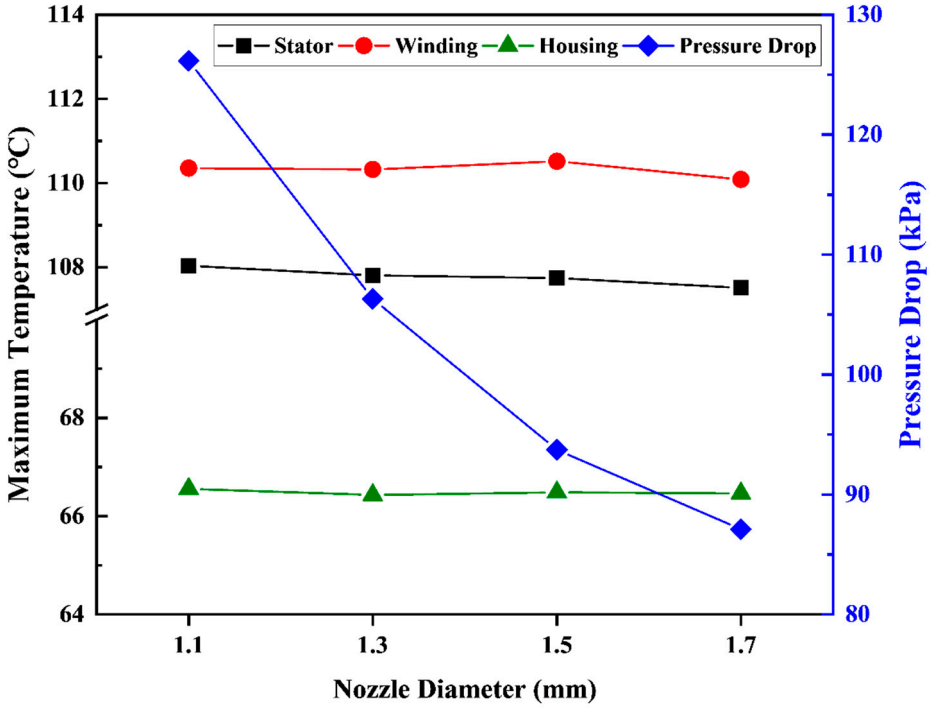


Figure 7. Variations in maximum temperatures of motor components and pressure drop with different nozzle diameters at Q_{in} = 20 LPM, T_{in} = 20 °C, and $\dot{Q}_{heat}$ = 5.10 kW.

In contrast to the limited thermal response, the pressure drop decreased monotonically from 126.17 kPa to 87.10 kPa as the D_n increased from 1.1 to 1.7 mm, corresponding to a 30.96% reduction. This behavior is attributed to the increased effective flow area and the

resulting decrease in coolant velocity and local hydraulic losses through the nozzles [41,47]. These findings demonstrate that nozzle diameter exerted a substantially greater influence on hydraulic performance than on the $T_{st,\,max}$, $T_{wd,\,max}$, and $T_{hs,\,max}$. Therefore, the 1.7 mm nozzle diameter was selected as the reference geometry for the subsequent flow rate and inlet temperature analyses, as it provided the lowest pressure drop while maintaining comparable thermal performance.

Figure 8 compares the stator, winding, and housing temperature fields obtained with nozzle diameters of 1.1–1.7 mm. Across the investigated nozzle diameters, the stator, winding, and housing exhibit broadly similar temperature distributions, indicating that nozzle diameter has only a limited effect on the overall thermal field. Localized hot spots remain concentrated near the upper and lower ends of the stator and winding, where heat generation and thermal resistance are comparatively high, whereas the housing maintains substantially lower temperatures due to direct contact with the indirect cooling channel. Increasing the nozzle diameter from 1.1 to 1.7 mm produces only minor changes in the extent and intensity of these hot regions, with no meaningful redistribution of the dominant thermal gradients. Thus, nozzle enlargement primarily improves hydraulic performance while producing only minor changes in the maximum component temperatures and spatial temperature patterns.

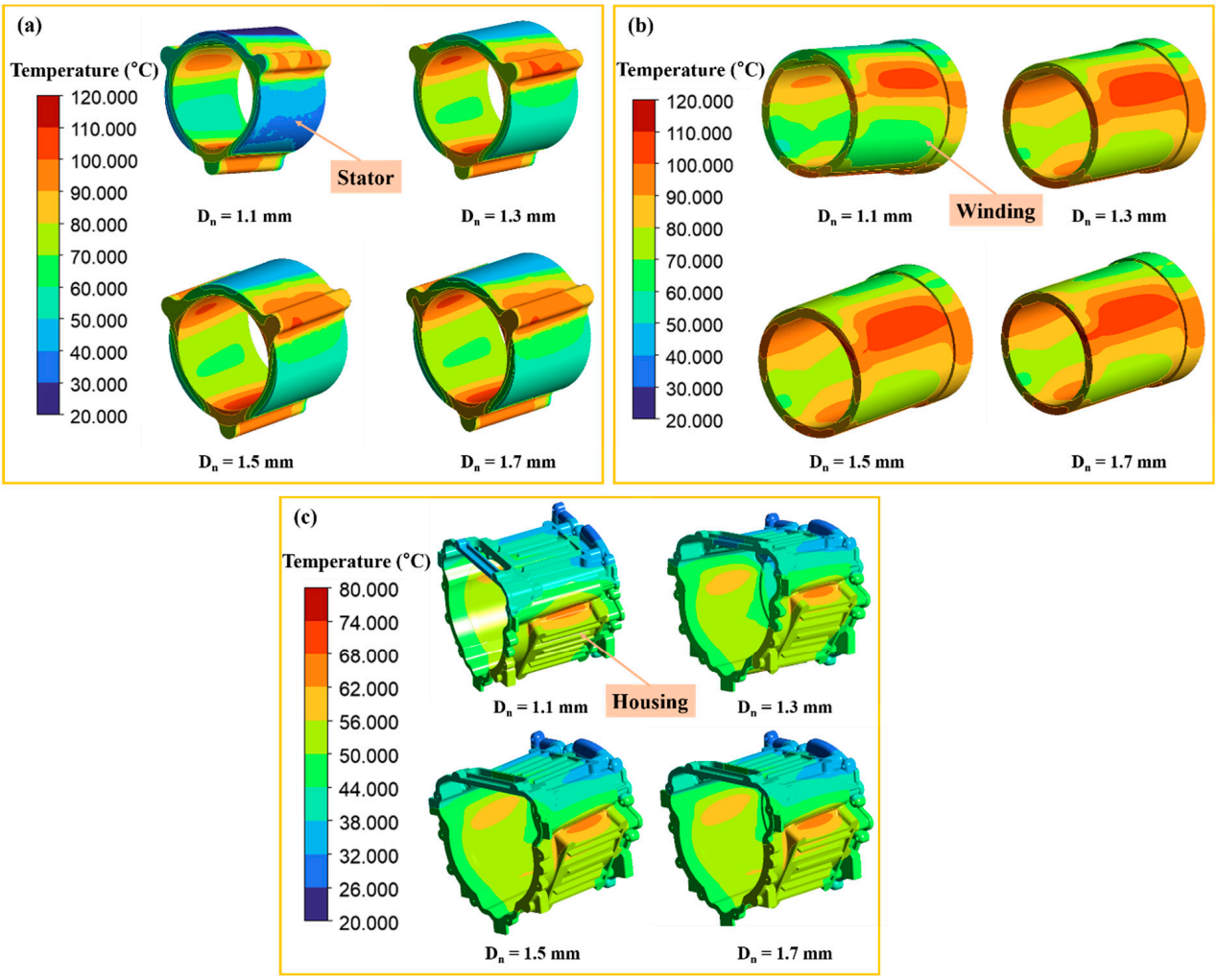


Figure 8. Temperature distributions of the (**a**) stator, (**b**) winding, and (**c**) housing at different nozzle diameters.

3.1.2. Effect of Volumetric Flow Rate (Q_{in})

Controlling the heat removal performance and hydraulic pressure drop of the CDICS relies heavily on the volumetric flow rate. Specifically, increasing flow rate enhances the heat capacity rate, promotes stronger forced convection, and reduces the thermal boundary-layer thickness, thereby reducing component temperatures and apparent convective thermal resistance and suppressing local hot spots. Conversely, insufficient flow limits coolant circulation and heat transport, resulting in higher component temperatures and potentially higher maximum temperatures and more extensive high-temperature regions. However, increasing the flow rate also intensifies frictional and local losses, leading to a higher pressure drop and greater pumping power demand. Therefore, the flow rate must be selected by balancing thermal improvements with hydraulic and energy penalties [17,41,47]. In this section, simulations were conducted at volumetric flow rates of 10, 12, 14, 16, 18, and 20 LPM to evaluate the corresponding thermal and hydraulic responses. Based on the nozzle diameter analysis, the nozzle diameter was set to 1.7 mm, and the inlet oil temperature was maintained at 20 °C.

Figure 9 presents the variations in the maximum temperatures of the stator, winding, and housing, together with the pressure drop across the cooling system, as a function of coolant volumetric flow rate. The maximum temperatures of all motor components decreased monotonically with increasing coolant volumetric flow rate. Elevating the coolant flow rate within the range of 10 to 20 LPM, the $T_{st,\,max}$, $T_{wd,\,max}$, and $T_{hs,\,max}$ decreased from 122.941 °C to 107.512 °C, 125.217 °C to 110.087 °C, and 80.496 °C to 66.465 °C, corresponding to reductions of 15.4 K, 15.1 K, and 14.0 K, respectively. An elevated coolant heat capacity rate, coupled with intensified forced-convection heat transfer, underpins this thermal enhancement [47,48]. The larger relative reduction in housing temperature reflects its direct thermal interaction with the indirect cooling channel.

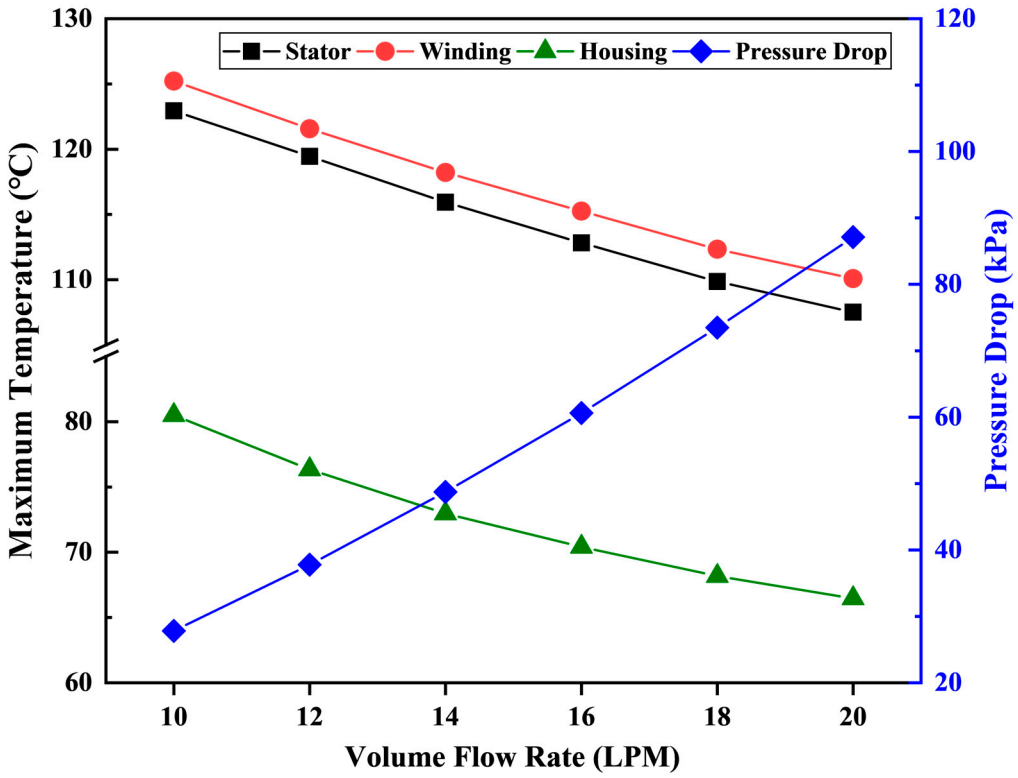


Figure 9. Variations in maximum temperatures of motor components and pressure drop with different volume flow rates at D_n = 1.70 mm, T_{in} = 20 °C, and $\dot{Q}_{heat}$ = 5.10 kW.

Conversely, the $\Delta P_{coolant}$ increased from 27.822 kPa to 87.10 kPa, representing a 213.06% increase, because higher coolant velocity intensified frictional and local losses within the nozzles and cooling passages [41,47]. Moreover, the progressively smaller temperature reductions at higher flow rates indicate diminishing thermal benefits, whereas the hydraulic penalty continued to increase. Since hydraulic pumping power is proportional to the product of pressure drop and volumetric flow rate, operation at 20 LPM provides the best thermal performance but also imposes the greatest auxiliary energy demand. These competing trends clearly demonstrate that selecting the operating condition based solely on thermal performance or hydraulic performance would lead to suboptimal system operation, thereby justifying the adoption of the subsequent multi-objective optimization framework based on ANN, NSGA-II, and TOPSIS.

Figure 10 shows that increasing the coolant volumetric flow rate progressively lowers the temperature levels of the stator, winding, and housing while preserving the general spatial pattern of the thermal field. At low flow rates, extensive high-temperature regions are observed in the stator and winding, particularly near the upper and lower ends and in areas adjacent to the principal heat-generating zones. As the flow rate increases, these hot regions contract and shift toward lower temperature bands, indicating reducing component temperatures and apparent convective thermal resistance. The housing remains considerably cooler than the internal components because of its direct interaction with the indirect cooling channel. Overall, higher flow rates reduced the maximum component temperatures and visually decreased the extent of the high-temperature regions. However, the thermal benefits of increasing volumetric flow rate are accompanied by a substantial hydraulic penalty. Higher flow rates intensify frictional and local losses in nozzles and cooling passages, thereby increasing the pressure drop and the corresponding pumping power requirement. Consequently, volumetric flow rate represents a critical decision variable in the subsequent multi-objective optimization, where improved thermal performance must be balanced against increased auxiliary energy consumption.

3.1.3. Effect of Inlet Oil Temperature (T_{in})

Inlet temperature is a key operating parameter because it governs the temperature difference between the coolant and the motor components, thereby directly influencing the heat transfer driving force and overall thermal level of the system [37]. To isolate its effect, the nozzle diameter and coolant volumetric flow rate were fixed at 1.7 mm and 15 LPM, respectively, based on the preceding analyses. Numerical simulations were conducted at inlet oil temperatures of 15, 17, 19, 21, 23, and 25 °C.

Figure 11 presents variations in the $T_{st,\,max}$, $T_{wd,\,max}$, and $T_{hs,\,max}$, along with the $\Delta P_{coolant}$, as the inlet oil temperature varies. Increasing T_{in} from 15 °C to 25 °C increased $T_{st,\,max}$, $T_{wd,\,max}$, and $T_{hs,\,max}$ from 109.772 °C to 118.851 °C, 112.086 °C to 121.087 °C, and 66.742 °C to 76.775 °C, respectively. Because the prescribed heat load remained fixed at 5.10 kW, these increases primarily represent an upward shift in the absolute component temperatures relative to the inlet coolant temperature, rather than a reduction in total heat rejection.

In addition, when T_{in} increased from 15 °C to 25 °C, the increases in stator, winding, and housing temperatures above the inlet changed from 94.772 K to 93.851 K, 97.086 K to 96.087 K, and 51.742 K to 51.775 K, respectively. These small differences indicate that the apparent thermal resistance remained nearly unchanged. Therefore, the principal effect was an approximately one-to-one upward displacement of the absolute temperature field, rather than a reduction in total heat rejection. Moreover, changes in the heat transfer coefficient cannot be inferred from maximum temperatures alone.

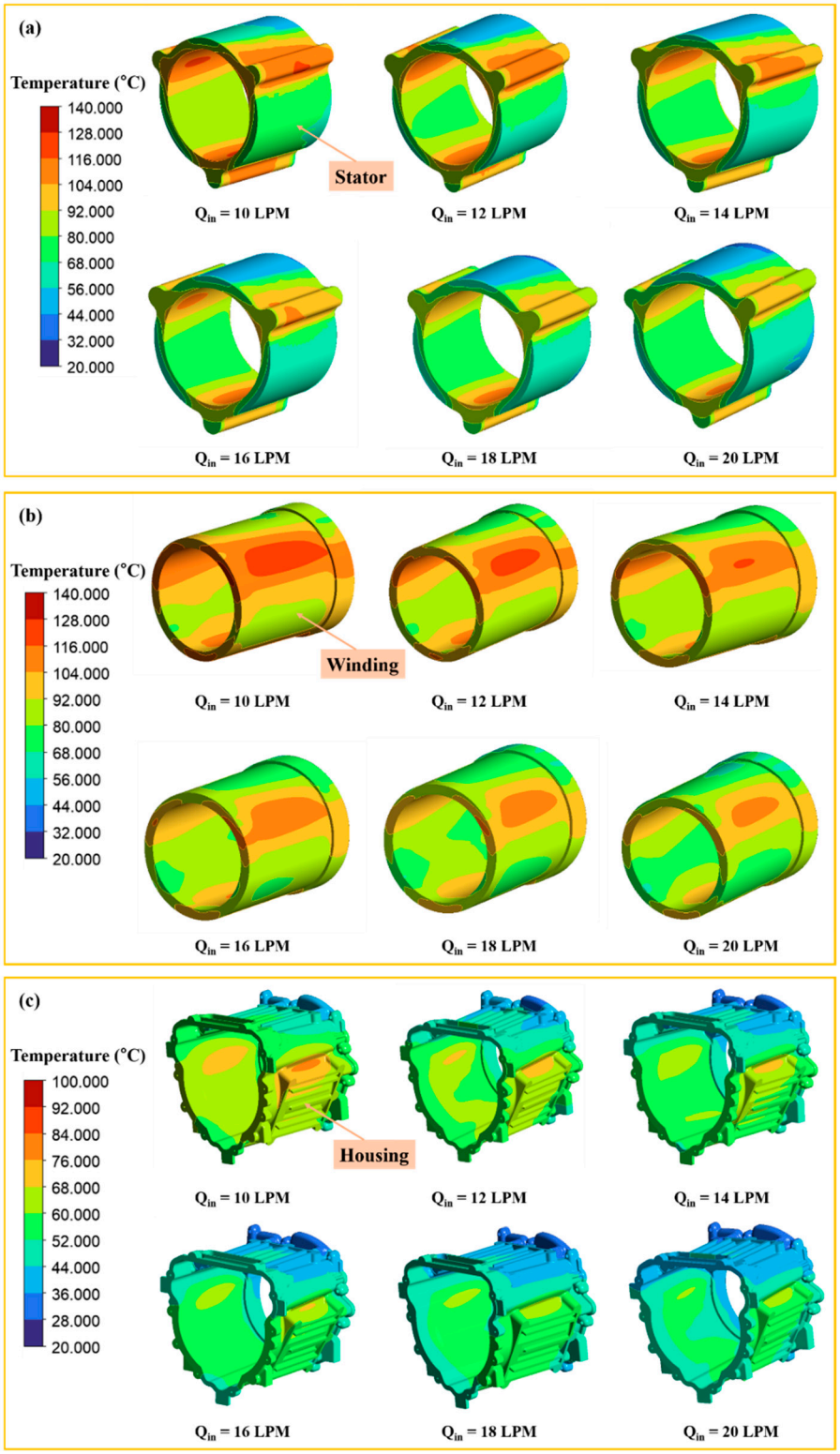


Figure 10. Temperature distributions of the (**a**) stator, (**b**) winding, and (**c**) housing at different volume flow rates.

Conversely, the $\Delta P_{coolant}$ decreased from 61.295 kPa to 47.956 kPa, representing a reduction of 21.76%. This decline is primarily due to the temperature-induced reduction in oil viscosity, which lowers frictional resistance in the nozzles and cooling passages. These opposing thermal and hydraulic trends demonstrate that inlet oil temperature is a critical optimization variable: lower temperatures improve thermal management but increase

pressure losses, whereas higher temperatures reduce hydraulic resistance at the expense of elevated component temperatures.

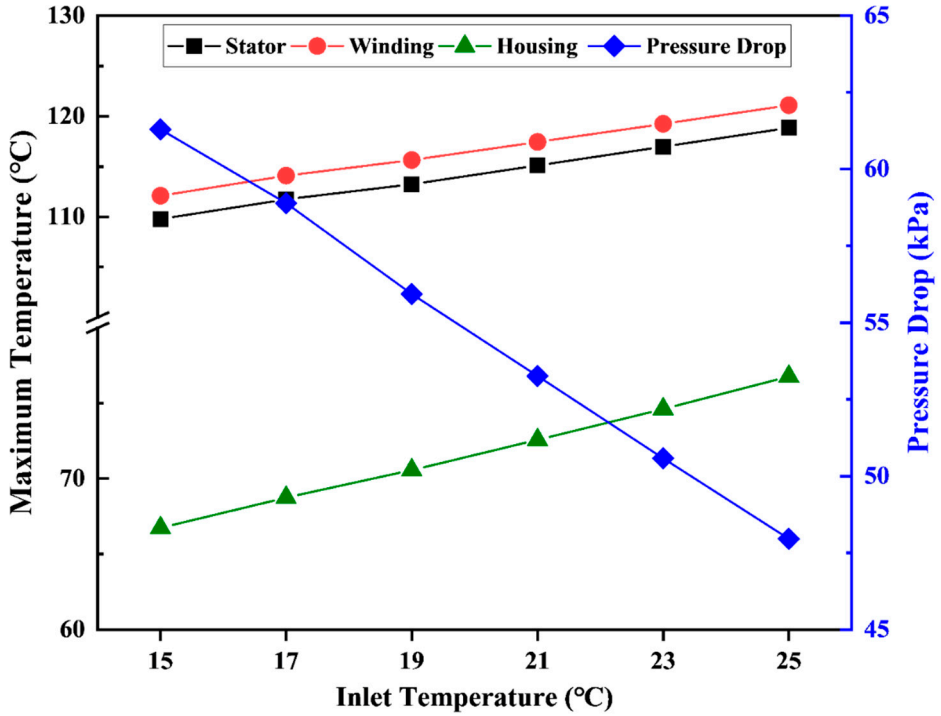


Figure 11. Variations in maximum temperatures of motor components and pressure drop with different inlet oil temperatures at D_n = 1.70 mm, Q_{in} = 15 LPM, and $\dot{Q}_{heat}$ = 5.10 kW.

Figure 12 shows a progressive upward shift in the absolute temperature fields as T_{in} increased. The stator and winding retain similar hotspot locations, primarily near the upper and lower end regions and the principal heat-generating zones. The housing also exhibits a systematic increase in surface temperature, although its absolute temperature remains lower due to direct heat exchange with the indirect cooling channel. Accordingly, these contours demonstrate the influence of coolant inlet temperature on the absolute thermal level but do not independently indicate changes in the heat transfer coefficient, as the hotspot locations and increases in temperature relative to the inlet remained similar.

Lowering T_{in} reduces the absolute component temperatures but does not increase the total steady-state heat rejection, which remains constrained by the prescribed 5.10 kW load and adiabatic external boundaries. However, colder oil has higher viscosity, which consequently increases the pressure drop and the ideal hydraulic pumping power. Thus, T_{in} represents a trade-off between absolute component temperature control and hydraulic demand. Therefore, the inlet oil temperature should be treated as a key decision variable in the subsequent thermo-hydraulic multi-objective optimization to balance thermal protection and pumping energy requirements.

3.2. Multi-Objective Optimization

Following the single-parameter analysis, the three design and operating variables, D_n, Q_{in}, and T_{in}, were evaluated using the surrogate-assisted multi-objective optimization procedure detailed in Section 3.2.1. The prescribed design space and corresponding CFD database are summarized in Tables 9 and 10, respectively.

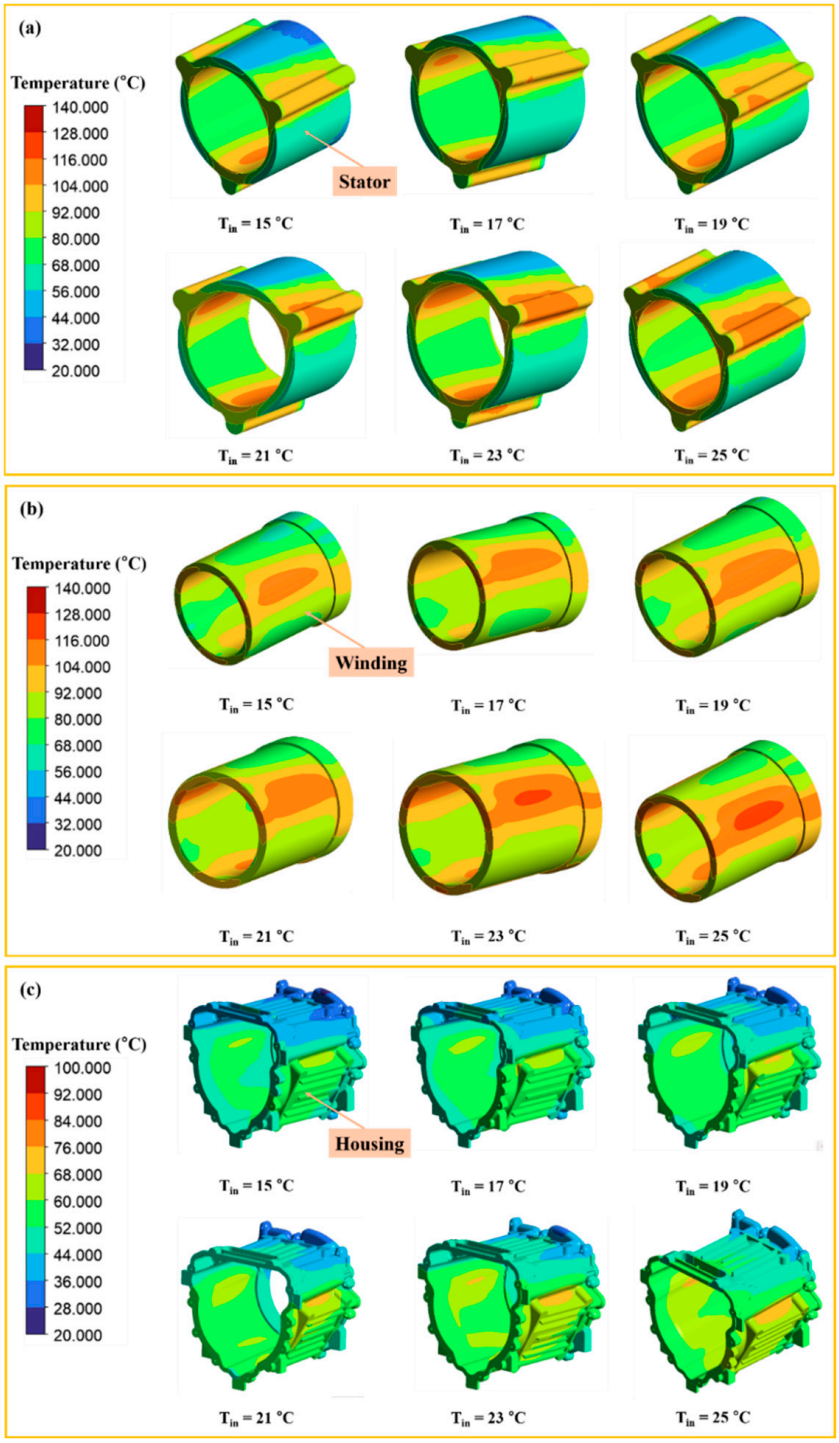


Figure 12. Temperature distributions of the (**a**) stator, (**b**) winding, and (**c**) housing at inlet oil temperatures.

Table 9. Parameter bounds used for the discrete-grid maximin space-filling design.

Parameter	Range	Units
Nozzle diameter (D_n)	1.1–1.7	mm
Volume flow rate (Q_{in})	10–20	LPM
Inlet oil temperature (T_{in})	15–25	°C

Table 10. Discrete-grid maximin design matrix and corresponding CFD thermo-hydraulic responses.

Case	D_n (mm)	Q_{in} (LPM)	T_{in} (°C)	$T_{st,\,max}$ (°C)	$T_{wd,\,max}$ (°C)	$T_{hs,\,max}$ (°C)	P_{pump} (W)
1	1.38	11.32	17.47	118.275	120.491	75.025	8.044
2	1.68	19.91	24.94	112.483	114.928	71.604	25.761
3	1.1	19.97	20.41	108.205	110.397	66.693	41.618
4	1.69	19.74	15.05	103.158	105.596	61.998	31.204
5	1.68	10.09	24.82	126.408	128.806	84.696	4.162
6	1.12	12.35	24.96	122.031	124.791	78.878	10.175
7	1.7	15.03	19.76	114.112	116.474	71.375	13.780
8	1.32	17.63	15.03	105.986	108.649	63.881	27.901
9	1.45	14.98	24.78	118.038	120.271	76.208	13.292
10	1.48	19.79	19.83	107.762	110.51	66.429	30.773
11	1.7	10.22	15.48	119.4	121.472	75.663	5.537
12	1.1	13.35	15.2	113.343	115.862	69.97	16.277
13	1.28	19.75	24.92	111.837	113.674	71.2	30.947
14	1.1	10.25	20.05	123.001	125.79	79.083	7.184
15	1.23	15.14	20.8	114.298	116.676	72.178	17.585
16	1.36	10.16	22.91	125.26	127.569	82.405	5.248
17	1.63	10.25	20.16	122.481	124.935	79.878	5.033
18	1.56	14.95	15.73	110.786	113.202	67.589	15.637
19	1.11	19.81	16.11	105.64	108.246	63.514	43.943
20	1.46	13.35	20.81	117.665	119.938	74.785	10.835
21	1.12	16.67	24.37	113.749	116.263	72.046	23.073
22	1.69	14.16	24.88	119.9	122.042	77.729	10.325
23	1.7	18.88	18.91	107.645	110.083	66.246	25.607
24	1.56	17.23	22.32	113.225	115.792	71.252	19.462
25	1.1	16.3	18.01	110.837	113.388	67.985	25.727
26	1.42	16.56	18.43	110.518	113.031	68.133	20.722
27	1.14	10.03	16.01	120.538	122.98	76.573	7.252
28	1.3	18.5	20.98	110.295	112.682	68.615	28.192
29	1.5	19.07	16.2	105.219	107.826	63.571	29.829
30	1.27	14.72	16.57	111.738	114.269	68.774	17.549

The design space in this study was defined by three input parameters: nozzle diameter (D_n), coolant volume flow rate (Q_{in}), and inlet oil temperature (T_{in}). These parameters were selected because they directly influence the thermo-hydraulic behavior of the cooling system. The nozzle diameter affects spray characteristics, jet velocity, local flow resistance, and oil distribution in the direct-cooling region. The volume flow rate determines the coolant heat capacity rate and convective heat transfer capability, but also governs the magnitude of frictional and local pressure losses in the cooling passages. The inlet oil temperature controls the temperature difference between the coolant and the heat-generating motor components, and it also affects oil viscosity and, therefore, hydraulic resistance. Accordingly, these three parameters are physically meaningful and strongly related to the system's thermal and hydraulic output responses.

The corresponding output responses considered in the optimization include the maximum stator temperature ($T_{st,\,max}$), maximum winding temperature ($T_{wd,\,max}$), maximum housing temperature ($T_{hs,\,max}$), and pumping power (P_{pump}). The first three responses represent the motor's thermal performance and cooling effectiveness, while the pumping power quantifies the auxiliary energy required to circulate the cooling oil. Therefore, minimizing the component temperatures and pumping power simultaneously provides a practical basis for evaluating the overall performance of the cooling system.

Table 9 details the specific limits and intervals of the investigated input parameters. The D_n was varied from 1.1 to 1.7 mm, the Q_{in} from 10 to 20 LPM, and the T_{in} from 15 to 25 °C. Within these prescribed ranges, 30 design points were generated using a discrete-grid maximin space-filling procedure. Each variable was discretized at a resolution of 0.01 within

the bounds reported in Table 9. Using a random seed of 30, a pool of 5000 candidate points was generated by independent uniform sampling from the discrete levels. The candidate variables were normalized using their prescribed minimum and maximum values. An initial point was selected using the seeded random-number generator, after which points were added sequentially by selecting the candidate with the greatest minimum Euclidean distance from the previously selected points. This maximin procedure promotes separation among points in the joint normalized design space but does not enforce Latin stratification within each marginal variable. For each sampled design point, CFD simulations were performed to obtain the corresponding thermal and hydraulic performance indices. The resulting design matrix and corresponding CFD responses are reported in Table 10. The resulting dataset was then used as the foundation for training an ANN-based surrogate model. This data-driven surrogate enables efficient prediction of system responses without requiring additional high-cost CFD simulations for every candidate design. Subsequently, the trained ANN model was coupled with NSGA-II to generate a final nondominated population, and TOPSIS was applied to select the most suitable compromise solution among the competing objectives.

3.2.1. Surrogate Modeling Using ANN and Optimization Framework

In this study, an integrated ANN-NSGA-II-TOPSIS framework was developed to perform multi-objective optimization of the CDICS for the PMSM electric motor. The ANN-NSGA-II-TOPSIS optimization framework included design-space definition, discrete-grid maximin space-filling CFD sampling, ANN surrogate modeling, NSGA-II Pareto optimization, and TOPSIS-based selection of a compromise solution. The framework was designed to overcome the high computational cost associated with direct CFD-based optimization. Instead of repeatedly performing CFD simulations for thousands of candidate designs, a data-driven surrogate model was first constructed using ANNs. The CFD dataset generated from the discrete-grid maximin space-filling design was used as the training database, where the input variables were nozzle diameter D_n, volume flow rate Q_{in}, and inlet oil temperature T_{in}, while the output responses were maximum stator temperature $T_{st,\,max}$, maximum winding temperature $T_{wd,\,max}$, maximum housing temperature $T_{hs,\,max}$, and pumping power P_{pump}. The full-precision CFD dataset and discrete-grid sampling code are provided in Supplementary Files S1 and S2, respectively.

Because the CFD database contained only 30 design points, the ANN models were treated as domain-bounded surrogates. Predictions supported by the sampled input region were interpreted as interpolation, whereas nominally feasible points lying outside the actual sampled support were explicitly identified as boundary extrapolation and subjected to independent CFD verification. A separate response-specific ANN model was constructed for each output. The dataset was divided into training and holdout subsets at a fixed 80:20 ratio, yielding 24 training cases and 6 holdout cases. The partitioning used a random seed of 42. Cases 9, 10, 16, 18, 24, and 28 constituted the holdout set, while the remaining 24 cases constituted the training set. The input and output variables were standardized using StandardScaler, with the scaling parameters determined exclusively from the training data and subsequently applied to the holdout data. Each surrogate was implemented as a multilayer perceptron regressor using the lbfgs solver. For each response, predictions from an ensemble of nine ANN models, each initialized with a different random seed, were averaged to reduce sensitivity to initial-weight selection. The ensemble seeds were 42, 59, 76, 93, 110, 127, 144, 161, and 178. These measures mitigate, but cannot completely eliminate, the risk of overfitting associated with the limited dataset. The Python implementation for ANN training and prediction is provided in Supplementary File S3.

The ANN hyperparameters were optimized using an Optuna multivariate Tree-structured Parzen Estimator sampler with a random seed of 42 and 300 trials for each response. The search space included one or two hidden layers with 4–48 neurons per layer, an activation function, a regularization coefficient, a convergence tolerance, and a maximum function-evaluation limit. Candidate models were evaluated on the 24-case training set using RepeatedKFold with five folds, ten repetitions, and a random state of 42. During every cross-validation fold, the input and output scalers were fitted exclusively using the fold-training data and then applied to the corresponding validation data. The minimized selection criterion was

$$J = \overline{RMSE}_{CV} + 0.10 s_{RMSE,CV} \quad (6)$$

where $\overline{RMSE}_{CV}$ and $s_{RMSE,CV}$ are the mean and standard deviation over the 50 validation folds, respectively. Because the same cross-validation results were used to select hyperparameters, they were not interpreted as independent evidence of generalization. The final response-specific architectures and optimized hyperparameters are summarized in Table 11.

Table 11. Final ANN architectures and optimized hyperparameters.

Response	Hidden-Layer Architecture (Neurons)	Activation Function	Regularization Parameter, α	Convergence Tolerance	Maximum Function Evaluations
$T_{st,\,max}$	48	tanh	0.00250448	2.61027×10^{-4}	100,000
$T_{wd,\,max}$	36	tanh	0.00268165	3.07518×10^{-4}	120,000
$T_{hs,\,max}$	48	tanh	0.0304782	5.14807×10^{-4}	80,000
P_{pump}	12, 40	tanh	0.00287336	9.34664×10^{-7}	20,000

Note: All ANN models used the L-BFGS solver with a maximum of 10,000 iterations. The response-specific convergence tolerances and maximum function-evaluation limits are reported in the table.

After the ANN surrogate models were validated, they were retrained using the full CFD dataset and used as objective-function evaluators in the NSGA-II optimization. The optimization problem was formulated to minimize four responses simultaneously: $T_{st,\,max}$, $T_{wd,\,max}$, $T_{hs,\,max}$, and P_{pump}. The design vector was defined as x = [D_n, Q_{in}, T_{in}], with D_n = 1.1–1.7 mm, Q_{in} = 10–20 LPM, and T_{in} = 15–25 °C. A discrete resolution of 0.01 was applied to each design variable to ensure practical manufacturability and reproducible operating settings. NSGA-II was executed with a random seed of 42, a population size of 100, and 200 generations. Simulated binary crossover was applied with a probability of p_c = 0.90 and a distribution index of η_c = 15, while polynomial mutation was applied independently to each design variable with a probability of p_m = 1/3 and a distribution index of η_m = 20. Integer random sampling, rounding repair, and duplicate elimination were used to maintain feasible solutions within the prescribed design space. At generation 200, non-dominated sorting of the final NSGA-II population yielded 100 mutually nondominated solutions. This finite set represents an algorithmic approximation of the Pareto front within the bounded ANN-surrogate design space; it is not a mathematically complete or globally exact Pareto front. This trade-off set balances component temperatures against ideal hydraulic pumping power.

Optimization stability was evaluated using ten independent NSGA-II runs with random seeds of 42, 59, 76, 93, 110, 127, 144, 161, 178, and 195. The trained ANN surrogate models and all optimization parameters were held constant; only the NSGA-II random seed was changed. For comparison across runs, the objective values were min-max normalized using common bounds derived from the pooled nondominated reference set. Hypervolume was calculated

using the normalized reference point (1.10, 1.10, 1.10, 1.10), while the inverted generational distance (IGD) was evaluated relative to the pooled nondominated set.

Finally, TOPSIS was applied to rank the computed nondominated solutions. Because all objectives were minimized, their predicted values were min-max normalized before calculating Euclidean distances to the ideal and nadir solutions. Equal objective weights (0.25 each) served as the baseline. To evaluate whether the three correlated temperature objectives disproportionately influenced selection, sensitivity analysis varied total thermal-to-hydraulic weights across 90:10, 75:25, 50:50, 25:75, and 10:90, with thermal weight divided equally among the three temperatures. A correlation-controlled analysis also combined the normalized temperatures into a single thermal index and weighted it equally against pumping power. The highest-scoring solution under each scheme was retained as a preference-dependent compromise, providing engineers and manufacturers with design alternatives suited to different priorities in thermal management and pumping power. The NSGA-II–TOPSIS implementation and exported validation, sensitivity, and optimization outputs are provided in Supplementary Files S3 and S4, respectively.

3.2.2. Optimized Results and Validation

Figure 13 presents the Pearson correlation matrix between the design variables and output responses used for the ANN-based surrogate modeling and subsequent multi-objective optimization. The nozzle diameter (D_n) exhibits very weak correlations with the maximum stator, winding, and housing temperatures, with correlation coefficients of 0.03, 0.02, and 0.08, respectively, indicating that its direct influence on the thermal responses is limited within the investigated design range. However, D_n shows a weak negative correlation with pumping power (P_{pump}), with a coefficient of −0.23, suggesting that a larger nozzle diameter slightly reduces hydraulic resistance. In contrast, the volume flow rate (Q_{in}) has a dominant influence on both thermal and hydraulic performance. It is strongly negatively correlated with $T_{st,\,max}$, $T_{wd,\,max}$, and $T_{hs,\,max}$, with coefficients of −0.89, −0.89, and −0.83, respectively, confirming that increasing flow rate substantially enhances heat removal. Nevertheless, Q_{in} shows a strong positive correlation with P_{pump}, with a coefficient of 0.93, reflecting the significant hydraulic penalty associated with higher coolant circulation. The inlet oil temperature T_{in} shows moderate positive correlations with the three temperature responses, particularly with $T_{hs,\,max}$, which has a coefficient of 0.56. This correlation occurs because the inlet temperature serves as the baseline thermal reference for the steady-state system, whereby an increase in T_{in} induces a corresponding upward shift in the absolute temperature field under the fixed thermal load. Meanwhile, the three temperature responses are highly correlated with each other, with coefficients of 0.99–1.00, indicating strongly coupled thermal behavior among the motor components. Overall, the correlation results confirm the competing relationship between thermal performance and pumping power, thereby justifying the use of multi-objective optimization rather than single-objective parameter selection.

The agreement between the CFD results and ANN surrogate predictions was evaluated using the coefficient of determination (R^2) and root mean square error (RMSE), as summarized in Table 12. The training R^2 values ranged from 0.9993 to 1.0000, while the six-case holdout set yielded R^2 values of 0.9976–0.9994. The corresponding holdout RMSE values were 0.2869 K, 0.1941 K, 0.1835 K, and 0.2145 W for $T_{st,\,max}$, $T_{wd,\,max}$, $T_{hs,\,max}$, and P_{pump}, respectively. These results demonstrate close agreement for the available CFD samples. Accordingly, the holdout results support predictive agreement for the available unseen cases but do not establish unrestricted generalization. Therefore, the models were used primarily within the sampled domain, while the selected point at T_{in} = 15.00 °C,

which lies 0.03 °C below the minimum sampled value, was treated as marginal boundary extrapolation and independently verified using CFD.

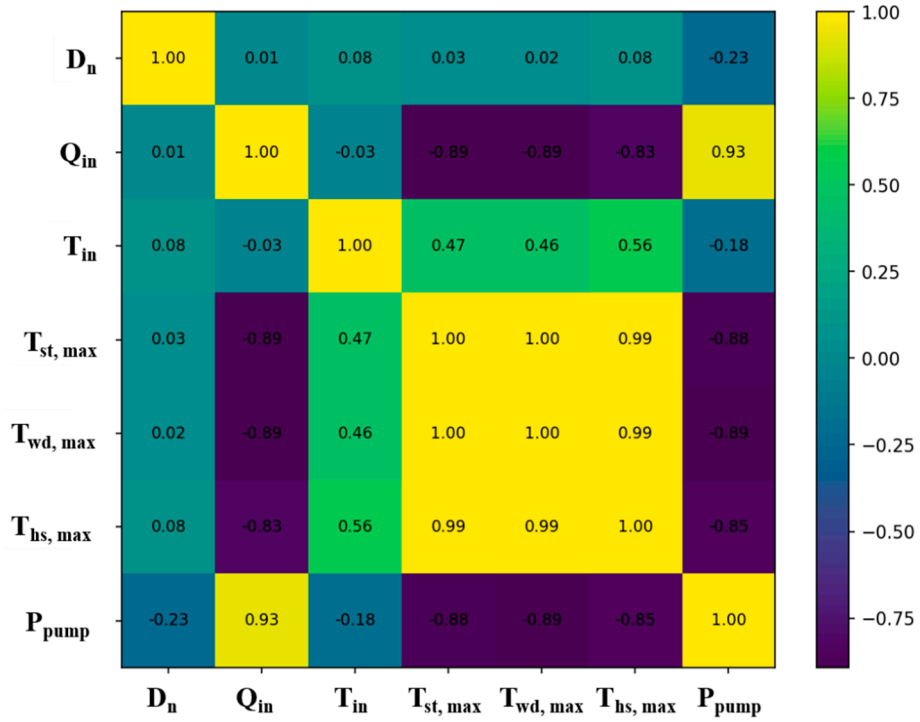


Figure 13. Correlation matrix of input design variables and CFD-derived thermo-hydraulic responses.

Table 12. ANN performance for the training and six-case holdout datasets.

	Training R^2	Holdout R^2	Training RMSE	Holdout RMSE
$T_{st, max}$	0.9997	0.9976	0.1058 K	0.2869 K
$T_{wd, max}$	0.9998	0.9989	0.0983 K	0.1941 K
$T_{hs, max}$	0.9993	0.9989	0.1586 K	0.1835 K
P_{pump}	1.0000	0.9994	0.0540 W	0.2145 W

The 80:20 data division yielded only six holdout cases. Therefore, the holdout R^2 values and Figure 14 were not interpreted as conclusive evidence of generalization or the absence of overfitting. Repeated cross-validation was used to improve the stability of hyperparameter selection across training-data partitions, whereas ensemble averaging over nine random initializations reduced sensitivity to stochastic model initialization and fitting. Neither procedure independently excludes overfitting, particularly because cross-validation is also involved in hyperparameter selection. Accordingly, the combined evidence supports using the ANN models as bounded interpolators within the sampled ranges of D_n, Q_{in}, and T_{in}, rather than as universally generalizable predictors. The subsequent CFD evaluation at the selected optimum serves only as a local consistency check and does not establish global surrogate robustness. Future work will expand the CFD database and incorporate independent validation and surrogate-uncertainty quantification.

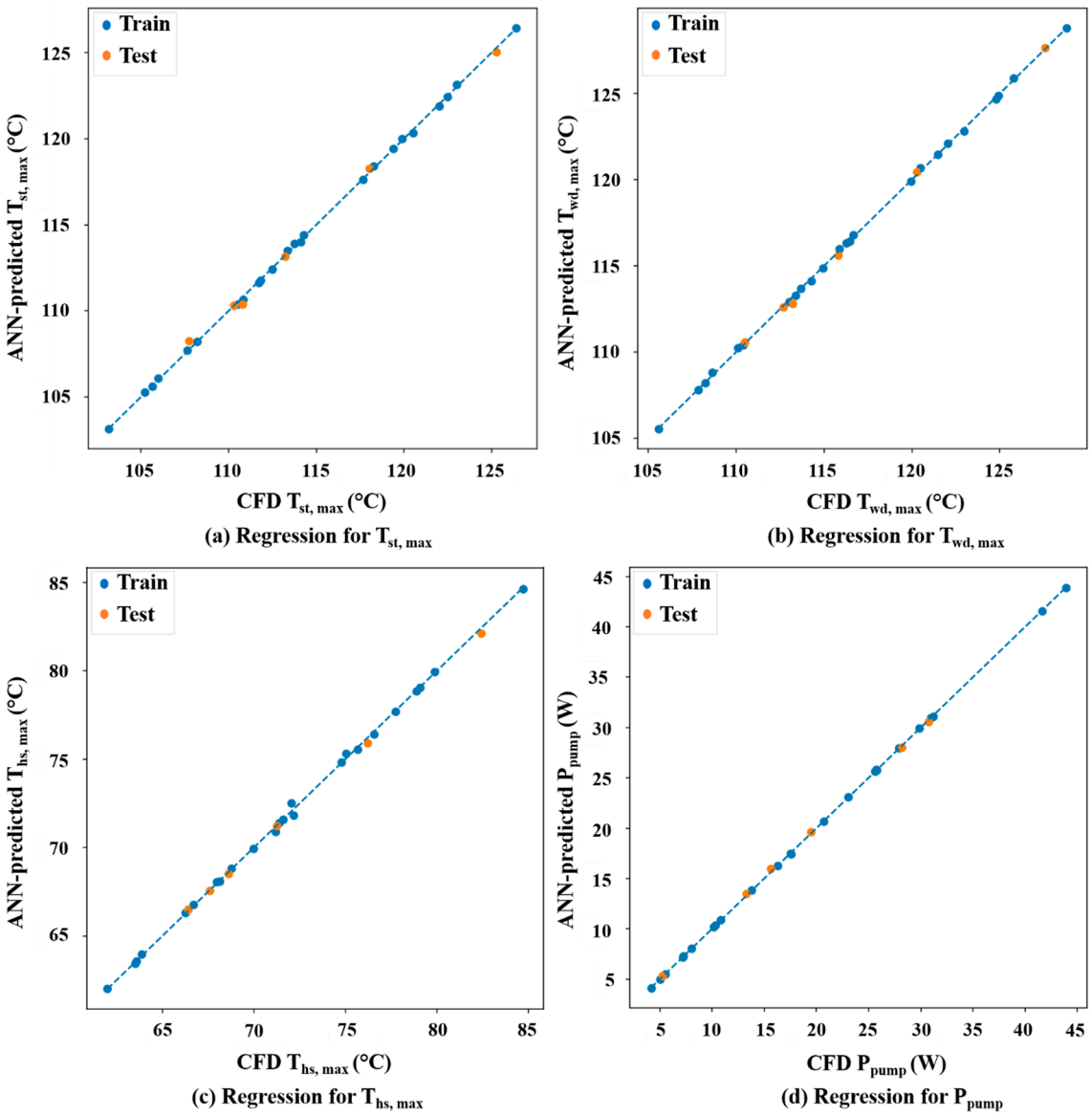


Figure 14. Comparison between CFD results and ANN predictions for evaluating interpolation accuracy within the sampled design space.

Figure 14 compares the CFD-derived and ANN-predicted values of $T_{st, max}$, $T_{wd, max}$, $T_{hs, max}$, and P_{pump}. The available training and holdout points lie close to the 45° reference line, indicating close agreement between the CFD results and ANN predictions for the sampled cases. No pronounced systematic bias is evident within these data. Nevertheless, predicted-versus-actual agreement on a six-case holdout set cannot independently rule out overfitting or demonstrate performance beyond the sampled parameter ranges. Therefore, Figure 14 supports the use of the ANN models for bounded interpolation during NSGA-II optimization within the investigated design space, rather than unrestricted predictive generalization.

To assess whether the complexity of the ANN surrogate was justified, its predictive performance was compared with that of a conventional second-order response surface model (RSM). A separate full quadratic model was fitted for each response:

$$\hat{y} = \beta_0 + \sum_{i=1}^{3} \beta_i x_i + \sum_{i=1}^{3} \beta_{ii} x_i^2 + \sum_{i<j} \beta_{ij} x_i x_j \quad (7)$$

where $x = [D_n, Q_{in}, T_{in}]$. The RSM contained an intercept, three linear terms, three quadratic terms, and three two-factor interaction terms and was fitted using ordinary least squares. For a direct comparison, the RSM and ANN used the identical 24-case training set and six-case holdout set comprising Cases 9, 10, 16, 18, 24, and 28. Repeated five-fold cross-validation with ten repetitions and a random seed of 42 was applied to the training set. During each cross-validation fold, preprocessing parameters were fitted exclusively using the corresponding fold-training data. The holdout cases were not used for model fitting, preprocessing, or model selection.

As shown in Table 13, the quadratic RSM achieved holdout R^2 values between 0.9933 and 0.9979, demonstrating that the principal relationships within the investigated design domain can be represented effectively by a simpler surrogate. Nevertheless, the ANN produced lower holdout RMSE for all four responses. Relative to the RSM, the ANN reduced the holdout RMSE by approximately 31.0%, 58.0%, 29.0%, and 69.9% for $T_{st,\,max}$, $T_{wd,\,max}$, $T_{hs,\,max}$, and P_{pump}, respectively. Although the RSM produced a marginally lower mean cross-validation RMSE for $T_{hs,\,max}$, the ANN performed better on its untouched holdout cases. Therefore, the ANN was retained for subsequent NSGA-II optimization because it consistently achieved superior independent holdout accuracy across all four objectives. These results remain subject to the limited size of the six-case holdout set.

Table 13. Comparison of ANN and second-order RSM performance using identical training and holdout datasets.

Response	Model	Training RMSE	Training R^2	Cross-Validation RMSE, Mean ± SD	Holdout RMSE	Holdout R^2
$T_{st,\,max}$	ANN	0.1058 K	0.9997	(0.5366 ± 0.1931) K	0.2869 K	0.9976
	RSM	0.3190 K	0.9974	(0.7205 ± 0.2933) K	0.4161 K	0.9950
$T_{wd,\,max}$	ANN	0.0983 K	0.9998	(0.5183 ± 0.1889) K	0.1941 K	0.9989
	RSM	0.3232 K	0.9973	(0.8015 ± 0.3320) K	0.4616 K	0.9936
$T_{hs,\,max}$	ANN	0.1586 K	0.9993	(0.3512 ± 0.1864) K	0.1835 K	0.9989
	RSM	0.1781 K	0.9991	(0.3409 ± 0.1257) K	0.2582 K	0.9979
P_{pump}	ANN	0.0540 W	1.0000	(0.4434 ± 0.6083) W	0.2145 W	0.9994
	RSM	0.3948 W	0.9988	(0.7542 ± 0.3121) W	0.7125 W	0.9933

To provide transparency regarding the available thermo-hydraulic trade-offs, Figure 15 presents the 100 nondominated solutions extracted from the final NSGA-II population (seed 42, generation 200) using a parallel-coordinate plot. This finite reported set approximates, but does not constitute, the mathematically complete Pareto front within the bounded surrogate-model design space. All axes were min-max normalized over the computed set. Each line represents one nondominated solution, and its color indicates the TOPSIS closeness coefficient under the equal-objective weighting scheme. The figure illustrates the relationships between D_n, Q_{in}, and T_{in} and the competing objectives $T_{st,\,max}$, $T_{wd,\,max}$, $T_{hs,\,max}$, and P_{pump}. Solution 16, highlighted in red, achieved the highest closeness coefficient under the predefined equal-objective decision rule and was therefore selected as the constrained compromise design within the investigated design space.

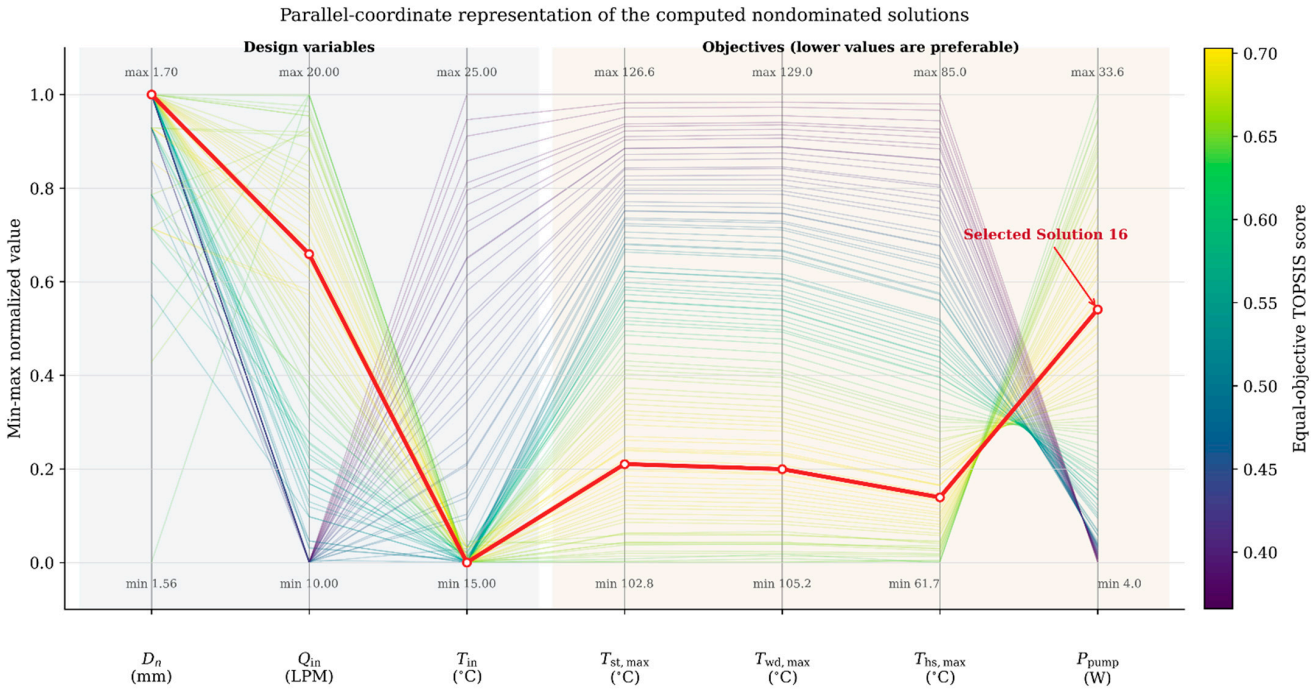


Figure 15. Parallel-coordinate representation of the 100 nondominated solutions extracted from the final NSGA-II population. Axes are normalized over this reported set; Solution 16 is highlighted in red.

The convergence and initialization sensitivity of NSGA-II were assessed using the ten independent runs summarized in Appendix A, Table A2. Figure 16 presents the individual normalized hypervolume trajectories along with their mean and standard deviation. The mean normalized hypervolume increased from (0.8156 ± 0.0024) at generation 25 to (0.8192 ± 0.0011) at generation 100 and remained effectively unchanged at (0.8193 ± 0.0011) at generation 200. The relative difference between the mean values at generations 100 and 200 was approximately 0.001%, while the final coefficient of variation among the ten runs was 0.134%. Moreover, the equal-objective TOPSIS solutions were confined to narrow ranges of D_n = 1.68–1.70 mm, Q_{in} = 16.23–16.69 LPM, and T_{in} = 15.00–15.01 °C. These results indicate that 200 generations were sufficient and that the selected compromise region was minimally sensitive to NSGA-II initialization.

As an additional optimizer-independent check, Table 14 compares the selected NSGA-II solution with the result of a deterministic Cartesian-grid search conducted using the trained surrogate models. The complete design domain was first evaluated using a 0.05 increment, comprising 13 × 201 × 201 = 525,213 design combinations. A subsequent local refinement was conducted at a resolution of 0.01 over D_n = 1.65–1.70 mm, Q_{in} = 15.50–17.50 LPM, and T_{in} = 15.00–15.10 °C. This refinement comprised 6 × 201 × 11 = 13,266 combinations. The refined candidates were combined with the coarse-grid nondominated solutions, after which nondominated sorting and the same equal-objective TOPSIS rule were reapplied. The deterministic search selected D_n = 1.70 mm, Q_{in} = 16.49 LPM, and T_{in} = 15.00 °C. This differs from the NSGA-II solution by only 0.10 LPM in flow rate, while the predicted responses differ by no more than 0.17 K for the motor's component temperature and 1.56% for ideal hydraulic pumping power. Therefore, the deterministic search independently confirmed the same boundary structure and compromise region identified by NSGA-II.

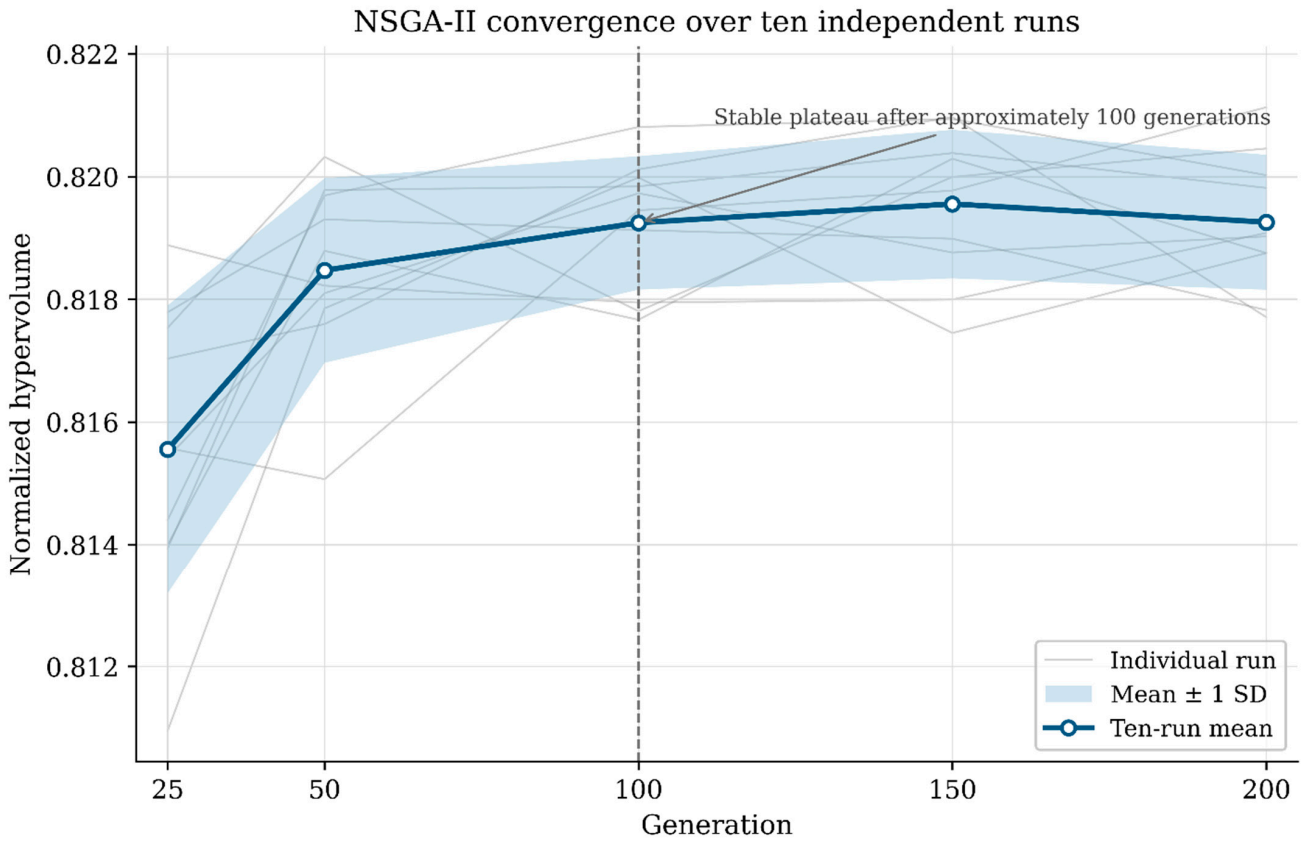


Figure 16. Normalized hypervolume convergence over ten independent NSGA-II runs. Thin gray lines represent individual runs, the solid line represents the ten-run mean, and the shaded region indicates ±1 standard deviation. The vertical dashed line marks generation 100, after which the normalized hypervolume reaches an approximately stable plateau.

Table 14. Comparison of the NSGA-II and deterministic-search compromise solutions.

Method	D_n (mm)	Q_{in} (LPM)	T_{in} (°C)	$T_{st,\,max}$ (°C)	$T_{wd,\,max}$ (°C)	$T_{hs,\,max}$ (°C)	P_{pump} (W)
NSGA-II–TOPSIS	1.70	16.59	15.00	107.7853	109.9406	64.9545	20.0133
Deterministic search	1.70	16.49	15.00	107.9558	110.1023	65.0699	19.7002
Absolute difference	-	-	-	0.17 K	0.16 K	0.12 K	0.31 W (1.56%)

In the baseline TOPSIS analysis, equal objective weights of 0.25 were assigned to $T_{wd,\,max}$, $T_{st,\,max}$, $T_{hs,\,max}$, and P_{pump}. The temperature objectives retain distinct engineering meanings related to winding insulation reliability, stator thermal loading, and housing heat accumulation. However, because they are strongly correlated, equal objective weighting gives the thermal group a combined weight of 0.75 compared with 0.25 for pumping power. Consequently, this scheme was treated as an equal-objective reference rather than a strictly balanced thermo-hydraulic weighting. Weighting sensitivity and grouped thermal analyses were therefore conducted to quantify how different application priorities affect the TOPSIS ranking. In the absence of manufacturer-defined priorities, the baseline solution is interpreted as a preference-dependent reference compromise, while the alternative schemes provide design options for different thermal reliability and auxiliary power requirements.

Table 15 summarizes the TOPSIS sensitivity analysis applied as post-processing to the same computed nondominated set. For each weighting scheme, the min-max-normalized objectives were weighted before calculating distances to the ideal and nadir solutions and the corresponding closeness coefficient. To control for correlation among the temperature

objectives, a grouped thermal index G_T was additionally evaluated against normalized pumping power using equal weights of 0.50, as follows:

$$G_T = \frac{z_{st} + z_{wd} + z_{hs}}{3} \tag{8}$$

where z_{st}, z_{wd}, and z_{hs} represent the normalized maximum temperatures of the stator, winding and housing, respectively.

Table 15. TOPSIS weighting-sensitivity results for the reported final-population nondominated solution set.

Weighting Scheme	Weights	Solution No.	D_n (mm)	Q_{in} (LPM)	T_{in} (°C)	$T_{st,\,max}$ (°C)	$T_{wd,\,max}$ (°C)	$T_{hs,\,max}$ (°C)	P_{pump} (W)	TOPSIS Closeness Coefficient
Thermal dominant 90:10	0.30, 0.30, 0.30, 0.10	63	1.70	18.85	15.00	104.2731	106.6132	62.7188	27.7506	0.8520
Equal-objective reference 75:25 (Selected design)	0.25, 0.25, 0.25, 0.25	16	1.70	16.59	15.00	107.7853	109.9406	64.9545	20.0133	0.7029
Group balanced 50:50	0.1667, 0.1667, 0.1667, 0.50	52	1.70	11.69	15.00	116.6659	118.6238	72.4908	8.0130	0.7272
Hydraulic priority 25:75	0.0833, 0.0833, 0.0833, 0.75	55	1.69	10.05	15.00	119.3324	121.4480	75.4909	5.3648	0.8764
Hydraulic dominant 10:90	0.0333, 0.0333, 0.0333, 0.90	33	1.70	10.00	19.38	122.2667	124.7014	79.6703	4.6647	0.9455
Grouped thermal index 50:50	0.50, 0.50	76	1.70	14.04	15.00	112.3797	114.3418	68.4195	12.9332	0.6697

Note: For the first five schemes, weights follow [$T_{st,\,max}$, $T_{wd,\,max}$, $T_{hs,\,max}$, P_{pump}]. The final row uses [G_T, P_{pump}], where $G_T = (z_{st} + z_{wd} + z_{hs})/3$. Because each weighting scheme defines a different decision preference, TOPSIS closeness coefficients should be compared only among solutions evaluated under the same scheme.

To ensure reproducibility of the TOPSIS procedure, each objective was normalized over the computed nondominated set using min-max normalization:

$$z_{ij} = \frac{f_{ij} - f_j^{min}}{f_j^{max} - f_j^{min}} \tag{9}$$

where f_{ij} is the value of objective j for solution i, and f_j^{min} and f_j^{max} are its minimum and maximum values, respectively. Because all four objectives were minimized, $z_{ij} = 0$ represents the ideal value, and $z_{ij} = 1$ represents the nadir value. For a specified weight vector w_j, the distances from the ideal and nadir solutions were calculated as

$$D_i^+ = \sqrt{\sum_{j=1}^{4} \left(w_j z_{ij}\right)^2} \tag{10}$$

$$D_i^- = \sqrt{\sum_{j=1}^{4} \left[w_j (1 - z_{ij})\right]^2} \tag{11}$$

The TOPSIS relative closeness coefficient was then determined from

$$C_i^* = \frac{D_i^-}{D_i^+ + D_i^-} \tag{12}$$

where a larger C_i^* indicates a solution closer to the ideal and farther from the nadir. In the objective order ($T_{st,\,max}$, $T_{wd,\,max}$, $T_{hs,\,max}$, P_{pump}), the raw ideal and nadir vectors of

the final-population nondominated solution set were f^+ = (102.7674, 105.1815, 61.71538, 3.984875) and f^- = (126.6255, 129.0441, 84.98381, 33.61764), respectively, where the temperature and pumping power components are expressed in °C and W. Because min-max normalization depends only on differences between values, it is invariant to an additive change in the temperature reference scale. Consequently, converting the temperature objectives from degrees Celsius to Kelvin produces identical normalized values, TOPSIS coefficients, and rankings.

The equal-objective scheme, assigning a weight of 0.25 to each response, was predefined as the primary decision rule. The remaining thermal-to-hydraulic weighting schemes and the grouped thermal-index case were evaluated only to determine how alternative engineering priorities affect the ranking. The ranking changed consistently with the specified priorities. Increasing thermal priority selected higher-flow designs with lower temperatures, whereas increasing hydraulic priority favored lower-flow designs with substantially lower pumping power. Under the primary equal-objective scheme, Solution 16 achieved the highest closeness coefficient (as shown in Appendix A, Table A1) and was therefore retained as the selected constrained compromise within the investigated design space. It corresponds to D_n = 1.70 mm, Q_{in} = 16.59 LPM, and T_{in} = 15.00 °C. Solution 76 was identified only under the supplementary grouped-criterion scenario and does not replace Solution 16 as the final design.

No universal component temperature constraints were imposed because allowable limits depend on the insulation system, magnet grade, lubricant, and service-life requirements. Nevertheless, the predicted temperatures remained below 180 °C, the applicable insulation-class limit where a verified Class H insulation system is employed. Therefore, Solution 16 represents a preference-selected constrained thermo-hydraulic compromise; application-specific qualification remains necessary.

Under the selected constrained compromise, the ANN surrogate model predicted maximum temperatures of 107.78 °C, 109.94 °C, and 64.95 °C for the stator, winding, and housing, respectively, with a corresponding pumping power of 20.01 W. The selected nozzle diameter is located at the upper bound of the design range, indicating that a larger nozzle is favorable for reducing hydraulic resistance while maintaining comparable cooling performance. Meanwhile, the optimized flow rate is lower than the maximum tested value of 20 LPM, demonstrating that the best compromise does not require the highest coolant flow rate. Instead, Q_{in} = 16.59 LPM maintains the selected component-temperature levels while avoiding excessive pumping power consumption. The inlet oil temperature was selected at the lower bound of 15.0 °C, reflecting the importance of maintaining a strong coolant-to-solid temperature difference to suppress the maximum temperatures of the motor components. Overall, the selected design represents a constrained thermo-hydraulic compromise that maintains numerically comparable maximum stator, winding, and housing temperatures within the assessed numerical and turbulence-model sensitivities while lowering the CFD-predicted ideal hydraulic pumping power relative to the specified reference baseline.

It should be noted that the selected solution is located at the upper bound of nozzle diameter and the lower bound of inlet oil temperature. Although 15.00 °C lies within the prescribed design bounds, it is 0.03 °C below the minimum sampled value of 15.03 °C. Therefore, this prediction represents marginal boundary extrapolation, supported by the independent CFD verification, rather than interpolation within the original database. Within the prescribed design space and under the predefined equal-objective TOPSIS decision rule, the optimization favored the upper nozzle diameter bound and the lower inlet temperature bound; however, this result does not establish that further improvement would occur beyond the investigated limits. Therefore, the present optimum should be interpreted as a constrained

optimum within the investigated practical limits, rather than a global optimum over an unlimited design domain. The nozzle diameter range was restricted by geometric packaging, manifold integration, manufacturability, and the need to preserve sufficient structural spacing in the cooling passage. Similarly, the lower inlet temperature bound was selected to represent a feasible coolant supply condition for an oil-based vehicle thermal management system, because further temperature reduction may require additional chiller energy and may affect oil viscosity and pumping requirements. Future work will extend the design range and include system-level constraints to determine whether further improvement is achievable beyond the present bounds.

Table 16 compares the ANN predictions with CFD results for the selected design and seven additional cases distributed across the Pareto region and boundary neighborhood. At the selected design, the absolute differences in $T_{st,\,max}$, $T_{wd,\,max}$, $T_{hs,\,max}$, and P_{pump} were 0.34 K, 0.12 K, <0.01 K, and 0.01 W, respectively. Across the seven additional cases, the corresponding maximum deviations were 1.48 K, 1.17 K, 0.58 K, and 0.22 W. The seven additional cases were excluded from ANN retraining and therefore constitute independent CFD response checks. Their agreement supports the consistency of ANN-CFD responses at independently evaluated points distributed across the relevant Pareto and boundary regions. However, these CFD comparisons do not constitute physical validation, independently establish Pareto optimality, or demonstrate global surrogate accuracy within or beyond the investigated design space.

Table 16. Independent ANN-CFD response checks across the Pareto region and boundary neighborhood.

Solution No.	Pareto-Region Category	D_n (mm)	Q_{in} (LPM)	T_{in} (°C)	Response	ANN Prediction	CFD Result	Absolute Differences
1	Hydraulic extreme	1.7	10	25	$T_{st,\,max}$ (°C) $T_{wd,\,max}$ (°C) $T_{hs,\,max}$ (°C) P_{pump} (W)	126.6255 129.0441 84.98381 3.984875	127.0797 129.4114 85.29573 4.03265	0.4542 0.3673 0.31192 0.047775
36	Low-power region	1.7	10	18.64	$T_{st,\,max}$ (°C) $T_{wd,\,max}$ (°C) $T_{hs,\,max}$ (°C) P_{pump} (W)	121.7385 124.1336 78.95287 4.76053	122.0238 124.2413 79.31917 4.809517	0.2853 0.1077 0.3663 0.048987
76	Intermediate front	1.7	14.04	15	$T_{st,\,max}$ (°C) $T_{wd,\,max}$ (°C) $T_{hs,\,max}$ (°C) P_{pump} (W)	112.3797 114.3418 68.41948 12.93318	110.9018 113.1675 67.83877 12.91072	1.4779 1.1743 0.58071 0.02246
12	Near selected design	1.7	16.32	15.31	$T_{st,\,max}$ (°C) $T_{wd,\,max}$ (°C) $T_{hs,\,max}$ (°C) P_{pump} (W)	108.4631 110.6165 65.53332 19.0378	108.0883 110.4056 65.56188 18.95224	0.3748 0.2109 0.02856 0.08556
16	Selected design	1.7	16.59	15	$T_{st,\,max}$ (°C) $T_{wd,\,max}$ (°C) $T_{hs,\,max}$ (°C) P_{pump} (W)	107.7853 109.9406 64.95447 20.01325	107.45 109.82 64.96 20.02	0.3353 0.1206 0.00553 0.00675
68	Diameter-side neighbor	1.69	16.77	15.01	$T_{st,\,max}$ (°C) $T_{wd,\,max}$ (°C) $T_{hs,\,max}$ (°C) P_{pump} (W)	107.4587 109.6581 64.74419 20.64105	107.0781 109.418 64.66532 20.46243	0.3806 0.2401 0.07887 0.17862
13	High-cooling region	1.7	18.35	15	$T_{st,\,max}$ (°C) $T_{wd,\,max}$ (°C) $T_{hs,\,max}$ (°C) P_{pump} (W)	104.9892 107.2922 63.15789 25.94381	104.9002 107.3106 63.20143 25.82687	0.089 0.0184 0.04354 0.11694
3	Thermal extreme	1.7	20	15	$T_{st,\,max}$ (°C) $T_{wd,\,max}$ (°C) $T_{hs,\,max}$ (°C) P_{pump} (W)	102.7674 105.1815 61.80547 32.03413	102.8863 105.2702 61.78893 32.25889	0.1189 0.0887 0.01654 0.22476

The role of CFD verification should be distinguished from that of the optimization checks. Agreement between ANN predictions and CFD results demonstrates response

accuracy at the independently evaluated designs, but it does not independently prove Pareto optimality. Within the prescribed surrogate-based design space, nondominance is supported by the final-population nondominated solution set, consistency across ten independent NSGA-II runs, and agreement with the deterministic grid search. Accordingly, Solution 16 is interpreted as a stable constrained compromise under the predefined equal-objective TOPSIS rule, rather than as a globally proven optimum beyond the investigated domain.

To compare the selected design with a consistent baseline, the reference design was defined as D_n = 1.70 mm, Q_{in} = 20.00 LPM, and T_{in} = 20 °C, corresponding to the reference geometry identified in Section 3.1.1. Table 17 compares this baseline with the selected design. The nominal differences in the maximum stator, winding, and housing temperatures were <0.1 K, 0.3 K, and 1.5 K, respectively. Because these differences were smaller than the corresponding 1.98–2.29 K variations observed in the single-point laminar-SST comparison and were comparable to the mesh-induced variations, all three temperature predictions were interpreted as numerically and model-wise comparable rather than established thermal improvements. At Q_{in} = 20.00 LPM, the reference baseline produced a CFD-predicted pressure drop of 87.1 kPa, corresponding to an ideal hydraulic pumping power of 29.0 W. The selected condition produced 72.4 kPa and 20.0 W. Thus, the selected design provided a CFD-estimated reduction of 9.0 W (31.0%) in ideal hydraulic pumping power relative to the specified reference baseline. This comparison concerns only the hydraulic quantity calculated as $\Delta P_{coolant} \cdot Q_{in}$. Because T_{in} decreased from 20 °C to 15 °C, the coolant-conditioning requirements of the two conditions differ. Pump efficiency, chiller power, heat exchanger operation, and coolant-conditioning energy were not evaluated. Therefore, the 31.0% reduction should not be interpreted as total system-level energy savings.

Table 17. Thermo-hydraulic performance comparison between the reference and selected designs.

Performance Indicator	Reference Design	Selected Design (CFD)	Nominal Difference, $X_{ref} - X_{sel}$
$T_{st,\,max}$ (°C)	107.5	107.5	<0.1 K
$T_{wd,\,max}$ (°C)	110.1	109.8	0.3 K
$T_{hs,\,max}$ (°C)	66.5	65.0	1.5 K
P_{pump} (W)	29.0	20.0	9.0 W (31.0%)

Note: The reference design is D_n = 1.70 mm, Q_{in} = 20.00 LPM, and T_{in} = 20 °C. The selected design is D_n = 1.70 mm, Q_{in} = 16.59 LPM, and T_{in} = 15 °C. P_{pump} denotes ideal hydraulic pumping power. Positive nominal differences indicate lower CFD-predicted values for the selected design relative to the reference design. For the three temperature responses, these differences do not constitute thermal improvements, as they are smaller than the corresponding variations in the laminar-SST model. Differences were calculated from the original CFD results before rounding. The reported 31.0% value denotes a CFD-estimated reduction in ideal hydraulic pumping power relative to the reference baseline; it does not represent total cooling system energy savings.

It should be noted that the experimental system did not include differential pressure sensors or separate flow meters in the direct and indirect cooling branches. Moreover, the installed FD-XC20R1 flow meter attachment was rated to 15 LPM and was used only to monitor the total inlet flow during the two thermal comparison tests. Therefore, the pressure drop, branch-flow distribution, and ideal hydraulic pumping power at the reference and optimized conditions remain CFD-predicted quantities. The reported 31.0% reduction should accordingly be interpreted as a numerically predicted reduction in ideal hydraulic pumping power under consistently applied CFD assumptions. Future work will employ an appropriately rated flow meter, differential pressure transducers, and branch-flow measurements at the reference and selected conditions.

To separate the effect of coolant flow rate from that of inlet temperature, an additional controlled comparison was conducted using CFD simulations. Both configurations used D_n = 1.70 mm and T_{in} = 15 °C, while Q_{in} was increased from 15.00 LPM to 16.59 LPM. Therefore, this comparison isolates the flow rate effect; it does not represent a nozzle diameter effect.

As shown in Table 18, at T_{in} = 15 °C, increasing Q_{in} from 15.00 LPM to 16.59 LPM lowered the maximum stator, winding, and housing temperatures by 2.3, 2.3, and 1.8 K, respectively, while increasing ideal hydraulic pumping power by 4.7 W (30.7%). Because D_n and T_{in} were held constant, this comparison isolates the effect of flow rate. It demonstrates lower absolute component temperatures at the cost of greater hydraulic demand, but does not imply increased total heat rejection under the prescribed 5.10 kW thermal load.

Table 18. Common-temperature CFD comparison isolating the effect of flow rate.

Performance Indicator	Q_{in} = 15.00 LPM	Q_{in} = 16.59 LPM	Nominal Change, $X_{16.59} - X_{15.00}$
$T_{st,\,max}$ (°C)	109.8	107.5	−2.3 K
$T_{wd,\,max}$ (°C)	112.1	109.8	−2.3 K
$T_{hs,\,max}$ (°C)	66.7	65.0	−1.8 K
P_{pump} (W)	15.3	20.0	+4.7 W (+30.7%)

Note: Both cases use D_n = 1.70 mm and T_{in} = 15 °C. P_{pump} denotes ideal hydraulic pumping power. Negative nominal change values denote reductions relative to the Q_{in} = 15.00 LPM case, whereas positive values denote increases. Differences were calculated from the original CFD results before rounding.

4. Conclusions

In this study, a combined direct–indirect cooling system (CDICS) for a permanent-magnet synchronous motor was evaluated using a surrogate-assisted multi-objective optimization framework to identify a constrained compromise between maximum component temperatures and ideal hydraulic pumping power. A three-dimensional CFD model was established to resolve conjugate heat transfer and oil–air two-phase flow within the proposed cooling configuration. The predicted temperature at the NTC sensor location was compared with two local winding temperature measurements obtained from motor-dynamometer experiments. The absolute differences were 0.37 K and 0.50 K, both within the estimated ±1.80 K sensor conversion uncertainty. Using this locally evaluated numerical model, the coupled effects of nozzle diameter, coolant volumetric flow rate, and inlet oil temperature were systematically investigated. Based on these analyses, an integrated ANN-NSGA-II-TOPSIS optimization framework was developed to identify the preference-selected constrained thermo-hydraulic compromise while balancing cooling effectiveness and ideal hydraulic pumping power. The main conclusions are summarized as follows:

(a) Increasing the nozzle diameter from 1.1 to 1.7 mm had only a minor effect on component temperatures, reducing the maximum stator, winding, and housing temperatures by 0.5 K, 0.3 K, and 0.1 K, respectively. However, the pressure drop decreased by 30.96%, indicating that nozzle enlargement mainly improves hydraulic performance under the investigated conditions.

(b) Increasing the coolant volume flow rate from 10 to 20 LPM significantly reduced the maximum stator, winding, and housing temperatures by 15.4 K, 15.1 K, and 14.0 K, respectively. However, the pressure drop increased by 213.06%, confirming a strong trade-off between cooling enhancement and hydraulic loss.

(c) Increasing the inlet oil temperature from 15 °C to 25 °C increased the maximum stator, winding, and housing temperatures by 9.1 K, 9.0 K, and 10.0 K, respectively, while

reducing pressure drop by 21.76%. Thus, inlet oil temperature has opposite effects on thermal protection and hydraulic resistance.

(d) Within the investigated design space, the ANN surrogate models showed high interpolation agreement with the available CFD data, with R^2 values of 0.9976–0.9994 for the six holdout cases. Coupling the bounded ANN surrogates with NSGA-II and TOPSIS identified D_n = 1.7 mm, Q_{in} = 16.59 LPM, and T_{in} = 15.0 °C as the selected compromise design. CFD evaluation at this point yielded absolute differences of 0.34 K, 0.12 K, <0.01 K, and 0.01 W for $T_{st,\,max}$, $T_{wd,\,max}$, $T_{hs,\,max}$, and P_{pump}, respectively, relative to the ANN predictions, providing local confirmation of the selected solution.

(e) Compared with the reconciled reference baseline (D_n = 1.70 mm, Q_{in} = 20.00 LPM, and T_{in} = 20 °C), the selected design decreased the CFD-predicted ideal hydraulic pumping power from 29.0 W to 20.0 W, corresponding to an estimated reduction of 9.0 W (31.0%). The nominal differences in the maximum stator, winding, and housing temperatures were <0.1 K, 0.3 K, and 1.5 K, respectively. Because these differences were smaller than the corresponding 1.98–2.29 K variations observed in the single-point laminar-SST comparison and were comparable to the mesh-induced variations, all three temperature predictions were considered numerically and model-wise comparable rather than established thermal improvements. Because coolant flow rate and inlet temperature both differed, and pump efficiency, chiller power, heat exchanger operation, and coolant-conditioning energy were excluded, the reported reduction applies only to ideal hydraulic pumping power and does not represent total cooling system energy savings.

Although the present results indicate that the selected design maintained comparable component temperature predictions while lowering the CFD-predicted ideal hydraulic pumping power under the investigated assumptions, its practical applicability should be interpreted within the limitations of the present numerical framework. The simulations were conducted under prescribed operating conditions and did not include transient vehicle drive cycles, speed-dependent electromagnetic and mechanical losses, rotor-induced flow effects, long-term oil degradation, oil aeration/foaming behavior, or vehicle-representative external thermal boundary conditions. The steady-state VOF model also excluded a separate wall-film formulation, surface tension, wall adhesion, and contact-angle effects. Therefore, microscopic wetting, film-thickness evolution, and contact-line dynamics were neither resolved nor experimentally validated. As a result, the proposed optimization framework provides a controlled assessment of stator-side thermo-hydraulic design rather than a complete vehicle-level validation. Future work will couple the cooling model with transient drive-cycle loss maps, rotating-domain flow simulations, long-term oil property evaluation, realistic external convection, radiation, and mounting-conduction boundaries, surface-tension and contact-angle characterization, wall-film modeling, and local oil-coverage measurements to confirm the system-level applicability of the optimized CDICS. Maximum component temperatures, pressure drop, branch-flow distribution, and pumping power remain experimentally unvalidated. Future work should include multi-location temperature measurements, differential pressure measurements, and branch-resolved flow measurements. In addition, a broader sensitivity and optimization study, including channel thickness, width, and number, as well as manifold topology, will be conducted in future work to further improve the full CDICS architecture.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/electronics15194556/s1, File S1: The full-precision 30-case CFD input–response dataset; File S2: The Python 3.11.9 code for discrete-grid maximin space-filling sampling; File S3: The Python 3.11.9 code for ANN training, prediction and NSGA-II-TOPSIS analysis,

including the seeds, scaling procedures, and model and optimization settings; File S4: The exported analysis outputs, including the ANN-RSM comparison, repeated NSGA-II statistics, grid-search results, TOPSIS sensitivity analysis, ANN-CFD verification, and 100-solution nondominated set.

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Data Availability Statement: The non-proprietary data and code supporting this study are provided in four Supplementary Files. The selected design and seven additional CFD checks are also reported in Table 16. The proprietary CAD geometry cannot be disclosed; however, all shareable dimensions and numerical settings are reported in the manuscript.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Table A1. Nondominated solutions extracted from the final NSGA-II population and corresponding TOPSIS closeness coefficients under the equal-objective weighting scheme (w = [0.25, 0.25, 0.25, 0.25]).

Solution No.	D_n (mm)	Q_{in} (LPM)	T_{in} (°C)	Predicted $T_{st,\,max}$ (°C)	Predicted $T_{wd,\,max}$ (°C)	Predicted $T_{hs,\,max}$ (°C)	Predicted P_{pump} (W)	TOPSIS Closeness Coefficient
1	1.7	10	25	126.6255	129.0441	84.98381	3.984875	0.366025
2	1.56	20	15	102.9513	105.601	61.71538	33.61764	0.631971
3	1.7	20	15	102.7674	105.1815	61.80547	32.03413	0.646435
4	1.7	14.43	15.03	111.6742	113.6605	67.84295	13.88342	0.667179
5	1.66	12.78	15.1	114.5742	116.5808	70.54777	10.23662	0.603692
6	1.66	15.81	15	108.9946	111.1883	65.84092	17.87254	0.699104
7	1.7	14.68	15.33	111.4051	113.432	67.72636	14.41745	0.669134
8	1.7	10	24.46	126.2045	128.6545	84.5092	4.047938	0.369899
9	1.7	10	17.79	121.1564	123.4979	78.13911	4.873485	0.450856
10	1.7	10.02	18.44	121.569	123.9494	78.72366	4.813912	0.441354
11	1.7	11.82	15	116.438	118.3894	72.25195	8.249195	0.561773
12	1.7	16.32	15.31	108.4631	110.6165	65.53332	19.0378	0.698642
13	1.7	18.35	15	104.9892	107.2922	63.15789	25.94381	0.681208
14	1.7	14.87	15	110.8429	112.856	67.16961	15.02949	0.681024
15	1.7	10.01	21.13	123.5664	126.0619	81.36016	4.458316	0.403527
16	1.7	16.59	15	107.7853	109.9406	64.95447	20.01325	0.702894
17	1.7	10.02	19.07	122.0131	124.4294	79.33239	4.731297	0.43192
18	1.68	16.07	15.01	108.6186	110.7841	65.55084	18.5263	0.701274
19	1.7	13.08	15.33	114.3599	116.32	70.28441	10.6625	0.608561
20	1.65	12.57	15.1	114.9108	116.928	70.90521	9.840202	0.595459
21	1.7	16.46	15.01	108.014	110.1586	65.11327	19.60228	0.70288
22	1.7	19.75	15.03	103.1019	105.5023	62.01844	31.07433	0.65167
23	1.63	20	15	102.8298	105.3773	61.73721	32.74478	0.640006
24	1.7	10.02	23.58	125.4812	127.9646	83.67547	4.175296	0.377331
25	1.7	17.28	15	106.6411	108.8565	64.19925	22.24765	0.698225

Table A1. *Cont.*

Solution No.	D_n (mm)	Q_{in} (LPM)	T_{in} (°C)	Predicted $T_{st, max}$ (°C)	Predicted $T_{wd, max}$ (°C)	Predicted $T_{hs, max}$ (°C)	Predicted P_{pump} (W)	TOPSIS Closeness Coefficient
26	1.66	13.84	15.03	112.5783	114.622	68.70282	12.61908	0.64846
27	1.7	17.87	15	105.7107	107.9756	63.60687	24.25401	0.68991
28	1.7	10	22.64	124.765	127.2708	82.83231	4.262946	0.386152
29	1.7	10	24.11	125.9293	128.396	84.19541	4.088963	0.372637
30	1.68	10	17.57	120.9622	123.2846	77.91118	4.941195	0.454846
31	1.7	15.05	15	110.5129	112.5392	66.91418	15.51052	0.685651
32	1.7	10.03	15.93	119.9379	122.1339	76.36408	5.176536	0.481068
33	1.7	10	19.38	122.2667	124.7014	79.67035	4.664696	0.426977
34	1.7	10	16.51	120.334	122.5812	76.94451	5.050392	0.470978
35	1.7	14.16	15	112.1566	114.1249	68.23159	13.22403	0.658466
36	1.7	10	18.64	121.7385	124.1336	78.95287	4.76053	0.437874
37	1.7	19.54	15	103.3459	105.7323	62.1559	30.30463	0.656283
38	1.7	10.01	23.11	125.123	127.6202	83.257	4.219092	0.381585
39	1.7	10	22.06	124.307	126.8149	82.27853	4.33254	0.392378
40	1.67	13.74	15.03	112.8097	114.8274	68.87807	12.3397	0.644018
41	1.7	12.31	15	115.5658	117.5001	71.36044	9.175298	0.582415
42	1.7	10	15.3	119.6199	121.766	75.8594	5.226505	0.49018
43	1.69	10	21.51	123.8816	126.3797	81.74385	4.416581	0.398586
44	1.7	10.3	15.06	119.0177	121.1041	75.10588	5.706682	0.503573
45	1.7	10.98	15	117.879	119.8879	73.80319	6.789194	0.528559
46	1.7	11.19	15.15	117.611	119.6156	73.5431	7.111369	0.533822
47	1.7	11.68	15.21	116.8039	118.7812	72.68879	7.951498	0.552034
48	1.69	12.23	15.22	115.7879	117.759	71.67595	9.001345	0.575474
49	1.69	19.11	15.34	104.1809	106.5763	62.77418	28.5917	0.663315
50	1.69	16.95	15.02	107.1674	109.3831	64.5538	21.21695	0.700641
51	1.69	17.21	15.15	106.8379	109.0836	64.38516	22.00689	0.697108
52	1.7	11.69	15	116.6659	118.6238	72.49084	8.01295	0.556423
53	1.7	17.98	15	105.5424	107.8163	63.50147	24.63698	0.688039
54	1.69	13.44	15	113.4444	115.4048	69.37673	11.57831	0.631256
55	1.69	10.05	15	119.3324	121.448	75.49088	5.364775	0.497163
56	1.7	10	22.28	124.4804	126.9883	82.48942	4.306068	0.389958
57	1.7	15.53	15.23	109.7953	111.877	66.45582	16.74892	0.691557
58	1.7	10.01	17.08	120.6767	122.9663	77.45299	4.984463	0.462228
59	1.7	10	20.21	122.8799	125.3502	80.48042	4.559595	0.41535
60	1.7	18.17	15.01	105.2634	107.5529	63.33135	25.29918	0.684567
61	1.7	10	20.1	122.7975	125.2637	80.37292	4.573392	0.41685
62	1.7	15.17	15	110.2939	112.3292	66.74701	15.83649	0.688454
63	1.7	18.85	15	104.2731	106.6132	62.7188	27.75061	0.671056
64	1.7	11.99	15	116.1377	118.0818	71.94086	8.564043	0.568859
65	1.7	10.46	15	118.7297	120.7918	74.76153	5.960456	0.509939
66	1.69	11.3	15.02	117.3016	119.2964	73.21102	7.349636	0.541254
67	1.7	12.01	15	116.1022	118.0455	71.90437	8.60153	0.569699
68	1.69	16.77	15.01	107.4587	109.6581	64.74419	20.64105	0.701843
69	1.7	15.29	15	110.0757	112.1204	66.58235	16.1667	0.691023
70	1.66	19.3	15	103.657	106.116	62.29963	29.76249	0.658121
71	1.7	14.74	15.03	111.1013	113.1077	67.38294	14.67744	0.676808
72	1.7	19.97	15	102.8042	105.2165	61.82777	31.9209	0.647071
73	1.7	13.69	15.03	113.049	114.9984	69.00616	12.09913	0.639999
74	1.7	17.47	15.11	106.4164	108.6546	64.09605	22.82912	0.695016
75	1.7	11.47	15	117.0476	119.0186	72.89658	7.621902	0.547526
76	1.7	14.04	15	112.3797	114.3418	68.41948	12.93318	0.654087
77	1.69	19.2	15	103.7901	106.1777	62.41422	29.1225	0.662823
78	1.7	17.75	15.06	105.9402	108.1987	63.77403	23.80815	0.691428
79	1.7	10	16.03	120.0432	122.2515	76.5084	5.119138	0.4786
80	1.7	19.54	15.45	103.709	106.1124	62.53352	30.03829	0.654586

Table A1. *Cont.*

Solution No.	D_n (mm)	Q_{in} (LPM)	T_{in} (°C)	Predicted $T_{st, max}$ (°C)	Predicted $T_{wd, max}$ (°C)	Predicted $T_{hs, max}$ (°C)	Predicted P_{pump} (W)	TOPSIS Closeness Coefficient
81	1.7	10.01	22.95	124.9961	127.4963	83.10688	4.238154	0.383144
82	1.7	13.21	15	113.9211	115.8542	69.78091	11.03119	0.620906
83	1.7	18.47	15	104.814	107.1262	63.04995	26.37343	0.678848
84	1.7	10.32	15.03	118.9694	121.0506	75.04346	5.741717	0.504727
85	1.7	10	19.67	122.4786	124.9269	79.953	4.627706	0.422836
86	1.67	12.43	15.11	115.2645	117.2534	71.19093	9.484811	0.58779
87	1.7	19.63	15.01	103.2382	105.6306	62.09414	30.63577	0.654311
88	1.7	10.01	16.39	120.2451	122.4814	76.81666	5.081594	0.473165
89	1.62	18.85	15	104.2447	106.7595	62.63765	28.45054	0.664186
90	1.67	19.2	15	103.7862	106.2183	62.3907	29.29642	0.66117
91	1.7	10.01	17.12	120.7023	122.9949	77.49028	4.978914	0.461597
92	1.67	10.46	15	118.598	120.6737	74.71291	6.028144	0.512061
93	1.69	12.54	15	115.0984	117.0463	70.9316	9.66345	0.593122
94	1.7	17.63	15.12	106.1715	108.4235	63.9433	23.36633	0.69277
95	1.7	16.76	15.07	107.5473	109.7222	64.82088	20.51927	0.701586
96	1.64	10.98	15	117.6126	119.6612	73.71295	6.945035	0.532894
97	1.69	10	21.49	123.8659	126.3637	81.72445	4.419021	0.398826
98	1.66	15.72	15.06	109.1932	111.3829	66.00725	17.58357	0.697204
99	1.7	16.28	15.31	108.5319	110.6819	65.58104	18.91598	0.698471
100	1.7	10.03	20.8	123.283	125.7678	81.00186	4.524731	0.408054

Note: The weights correspond to [$T_{st, max}$, $T_{wd, max}$, $T_{hs, max}$, P_{pump}]. Solution 16 achieved the highest closeness coefficient and was selected as the equal-objective compromise.

Table A2. Stability statistics from ten independent NSGA-II runs.

Stability Metric	Result
Number of independent runs	10
Random seeds	42, 59, 76, 93, 110, 127, 144, 161, 178, 195
Selected D_n, mean ± SD (range), mm	1.696 ± 0.007 (1.68–1.70)
Selected Q_{in}, mean ± SD (range), LPM	16.453 ± 0.162 (16.23–16.69)
Selected T_{in}, mean ± SD (range), °C	15.004 ± 0.005 (15.00–15.01)
Final normalized hypervolume, mean ± SD	0.8193 ± 0.0011
Hypervolume coefficient of variation	0.134%
Final IGD, mean ± SD	0.0080 ± 0.0004

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